JSS 1 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS

Learning Objectives

By the end of this lesson, students should be able to:

- Understand what whole numbers are
- Identify place values in whole numbers
- Order whole numbers from smallest to largest
- Understand properties of whole numbers

WHAT ARE WHOLE NUMBERS?

Whole numbers are counting numbers starting from zero. They include: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12... and so on. Whole numbers do not include fractions, decimals, or negative numbers.

Place Value

In our number system, the position of each digit determines its value. Consider the number 5,432:

Thousands	**Hundreds**	**Tens**	**Ones**
5	4	3	2

- 5 is in the thousands place = 5,000
- 4 is in the hundreds place = 400
- 3 is in the tens place = 30
- 2 is in the ones place = 2

So, 5,432 = 5,000 + 400 + 30 + 2

For larger numbers like 847,265:

Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
8	4	7	2	6	5

ORDERING WHOLE NUMBERS

To arrange whole numbers in order:

1. Compare the digits starting from the left
2. The number with the larger leftmost digit is greater

Example: Arrange in ascending order: 234, 567, 89, 1,234, 456

- First, look at how many digits each has
- 89 (2 digits) is smallest
- 234, 567, 456 (3 digits each)
- 1,234 (4 digits) is largest

Answer: 89, 234, 456, 567, 1,234

PROPERTIES OF WHOLE NUMBERS

1. Closure Property

- Addition: When you add two whole numbers, you get a whole number (5 + 3 = 8)
- Multiplication: When you multiply two whole numbers, you get a whole number (4 × 6 = 24)

2. Commutative Property

- Addition: a + b = b + a (Example: 5 + 7 = 7 + 5 = 12)
- Multiplication: a × b = b × a (Example: 3 × 4 = 4 × 3 = 12)

3. Associative Property

- Addition: $(a + b) + c = a + (b + c)$ (Example: $(2 + 3) + 4 = 2 + (3 + 4) = 9$)
- Multiplication: $(a \times b) \times c = a \times (b \times c)$ (Example: $(2 \times 3) \times 4 = 2 \times (3 \times 4) = 24$)

4. Identity Property

- Addition: $a + 0 = a$ (Adding zero doesn't change the number)
- Multiplication: $a \times 1 = a$ (Multiplying by 1 doesn't change the number)

PRACTICE QUESTIONS

1. Write the place value of each digit in 64,829
2. Arrange in descending order: 1,245, 987, 2,301, 156, 1,987
3. What is the value of 7 in 47,652?
4. Write the number in expanded form: 8,456

WEEK 2: LEAST COMMON MULTIPLE (LCM)

Learning Objectives

By the end of this lesson, students should be able to:

- Define multiples of a number
- Find the LCM of two or more numbers using the listing method
- Find the LCM using prime factorization

WHAT IS A MULTIPLE?

A multiple of a number is the product of that number and any whole number.

Example: Multiples of 3 are: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30... (3×1=3, 3×2=6, 3×3=9, and so on)

What is LCM?

The Least Common Multiple (LCM) is the smallest number that is a multiple of two or more numbers.

Method 1: Listing Multiples

Example 1: Find the LCM of 4 and 6

Step 1: List the multiples of each number

- Multiples of 4: 4, 8, **12**, 16, 20, **24**, 28, 32, **36**...
- Multiples of 6: 6, **12**, 18, **24**, 30, **36**...

Step 2: Identify common multiples: 12, 24, 36...

Step 3: The smallest common multiple is **12**

Therefore, LCM of 4 and 6 = 12

Example 2: Find the LCM of 3, 4, and 5

- Multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, **60**...
- Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, **60**...
- Multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, **60**...

LCM = **60**

Method 2: Prime Factorization Method

Example 1: Find the LCM of 12 and 18

Step 1: Write prime factorization of each number

- $12 = 2 \times 2 \times 3 = 2^2 \times 3$
- $18 = 2 \times 3 \times 3 = 2 \times 3^2$

Step 2: Take the highest power of each prime factor

- Highest power of $2 = 2^2$
- Highest power of $3 = 3^2$

Step 3: Multiply these together LCM = $2^2 \times 3^2 = 4 \times 9 =$ **36**

Example 2: Find the LCM of 8, 12, and 15

Step 1: Prime factorization

- $8 = 2 \times 2 \times 2 = 2^3$
- $12 = 2 \times 2 \times 3 = 2^2 \times 3$
- $15 = 3 \times 5$

Step 2: Highest powers

- Highest power of $2 = 2^3$
- Highest power of $3 = 3^1$
- Highest power of $5 = 5^1$

Step 3: Multiply LCM = $2^3 \times 3 \times 5 = 8 \times 3 \times 5 =$ **120**

EVALUATION

1. Find the LCM of 6 and 8
2. What is the LCM of 5 and 7?
3. Find the LCM of 4, 6, and 9
4. List the first five multiples of 12
5. Find the LCM of 10 and 15 using prime factorization

ASSIGNMENT

1. Find the LCM of 9 and 12
2. Calculate the LCM of 6, 8, and 12
3. What is the smallest number that is divisible by both 4 and 10?
4. Find the LCM of 7, 14, and 21
5. Use prime factorization to find the LCM of 18 and 24

WEEK 3: HIGHEST COMMON FACTOR (HCF)

Learning Objectives

By the end of this lesson, students should be able to:

- Define factors of a number
- Find the HCF of two or more numbers using the listing method
- Find the HCF using prime factorization

WHAT IS A FACTOR?

A factor of a number is a whole number that divides exactly into that number without leaving a remainder.

Example: Factors of 12 are: 1, 2, 3, 4, 6, 12 (Because 12÷1=12, 12÷2=6, 12÷3=4, 12÷4=3, 12÷6=2, 12÷12=1)

WHAT IS HCF?

The Highest Common Factor (HCF) is the largest number that divides exactly into two or more numbers.

Method 1: Listing Factors

Example 1: Find the HCF of 12 and 18

Step 1: List all factors of each number

- Factors of 12: 1, 2, 3, 4, **6**, 12
- Factors of 18: 1, 2, 3, **6**, 9, 18

Step 2: Identify common factors: 1, 2, 3, 6

Step 3: The highest common factor is **6**

Therefore, HCF of 12 and 18 = 6

Example 2: Find the HCF of 24, 36, and 48

- Factors of 24: 1, 2, 3, 4, 6, 8, **12**, 24
- Factors of 36: 1, 2, 3, 4, 6, 9, **12**, 18, 36
- Factors of 48: 1, 2, 3, 4, 6, 8, **12**, 16, 24, 48

Common factors: 1, 2, 3, 4, 6, 12

HCF = **12**

Method 2: Prime Factorization Method

Example 1: Find the HCF of 24 and 36

Step 1: Write prime factorization

- $24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$
- $36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Step 2: Take the lowest power of each common prime factor

- Lowest power of 2 = 2^2
- Lowest power of 3 = 3^1

Step 3: Multiply these together HCF = $2^2 \times 3 = 4 \times 3 =$ **12**

Example 2: Find the HCF of 30, 45, and 60

Step 1: Prime factorization

- $30 = 2 \times 3 \times 5$
- $45 = 3 \times 3 \times 5 = 3^2 \times 5$
- $60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Step 2: Common prime factors with lowest powers

- Lowest power of 3 = 3^1
- Lowest power of 5 = 5^1
- (Note: 2 is not in all three, so we don't include it)

Step 3: Multiply HCF = 3×5 = **15**

EVALUATION

1. Find the HCF of 8 and 12
2. List all factors of 20
3. What is the HCF of 16 and 24?
4. Find the HCF of 18, 27, and 36
5. Use prime factorization to find the HCF of 40 and 60

ASSIGNMENT

1. Find the HCF of 15 and 25
2. Calculate the HCF of 12, 18, and 24
3. What is the largest number that divides both 32 and 48?
4. Find the HCF of 20, 30, and 40
5. Use prime factorization to find the HCF of 54 and 72

WEEK 4: COUNTING IN BASE 2 (BINARY NUMBER SYSTEM)

Learning Objectives

By the end of this lesson, students should be able to:

- Understand the binary number system
- Count in base 2
- Differentiate between base 10 and base 2

INTRODUCTION TO NUMBER BASES

We normally use the base 10 (decimal) system, which has 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

The binary system (base 2) uses only 2 digits: **0 and 1**

Place Values in Base 2

In base 10, place values are powers of 10: 1, 10, 100, 1000... In base 2, place values are powers of 2: 1, 2, 4, 8, 16, 32, 64, 128...

128	**64**	**32**	**16**	**8**	**4**	**2**	**1**
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Counting in Base 2

Base 10	**Base 2**	**Explanation**
0	0	Zero
1	1	One
2	10	One 2, zero 1s
3	11	One 2, one 1

4	100	One 4, zero 2s, zero 1s
5	101	One 4, zero 2s, one 1
6	110	One 4, one 2, zero 1s
7	111	One 4, one 2, one 1
8	1000	One 8, zero 4s, zero 2s, zero 1s
9	1001	One 8, zero 4s, zero 2s, one 1
10	1010	One 8, zero 4s, one 2, zero 1s

Understanding Binary Numbers

Example 1: What does 1011_2 mean?

8	**4**	**2**	**1**
1	0	1	1

$1011_2 = (1\times8) + (0\times4) + (1\times2) + (1\times1) = 8 + 0 + 2 + 1 = 11_{10}$

Example 2: What does 1101_2 mean?

8	**4**	**2**	**1**
1	1	0	1

$1101_2 = (1\times8) + (1\times4) + (0\times2) + (1\times1) = 8 + 4 + 0 + 1 = 13_{10}$

Rules for Counting in Base 2

1. Start with 0
2. Next is 1
3. When you reach the highest digit (1), add 1 to the left and start over with 0

4. Continue this pattern

EVALUATION

1. Count from 0 to 10 in base 2
2. What comes after 111_2 in binary?
3. Convert 1010_2 to base 10
4. What is the place value of each digit in binary?
5. Write 15 in base 2

ASSIGNMENT

1. Count from 11_{10} to 16_{10} in base 2
2. What does 10101_2 equal in base 10?
3. Fill in the missing binary numbers: 100_2, 101_2, $\_\_\__2$, 111_2, $\_\_\__2$
4. Convert 1111_2 to base 10
5. What is the binary representation of 20_{10}?

WEEK 6: CONVERSION OF NUMBERS (BASE 10 TO BASE 2)

Learning Objectives

By the end of this lesson, students should be able to:

- Convert base 10 numbers to base 2
- Use the division method for conversion
- Verify their conversions

CONTENT

Method: Division by 2 (Repeated Division Method)

To convert a base 10 number to base 2:

1. Divide the number by 2
2. Write down the remainder (0 or 1)
3. Divide the quotient by 2 again
4. Continue until the quotient is 0
5. Read the remainders from bottom to top

Worked Examples

Example 1: Convert 13_{10} to base 2

$13 \div 2 = 6$ remainder 1 ↑

$6 \div 2 = 3$ remainder 0 |

$3 \div 2 = 1$ remainder 1 |

$1 \div 2 = 0$ remainder 1 | Read upwards

Reading from bottom to top: $\mathbf{1101_2}$

Verification: $1101_2 = (1\times8) + (1\times4) + (0\times2) + (1\times1) = 8 + 4 + 0 + 1 = 13_{10}$ ✓

Example 2: Convert 25_{10} to base 2

$25 \div 2 = 12$ remainder 1 ↑

$12 \div 2 = 6$ remainder 0 |

$6 \div 2 = 3$ remainder 0 |

$3 \div 2 = 1$ remainder 1 |

$1 \div 2 = 0$ remainder 1 | Read upwards

Answer: **11001_2**

Verification: $11001_2 = (1\times16) + (1\times8) + (0\times4) + (0\times2) + (1\times1) = 16 + 8 + 0 + 0 + 1 = 25_{10}$ ✓

Example 3: Convert 50_{10} to base 2

$50 \div 2 = 25$ remainder 0 ↑

$25 \div 2 = 12$ remainder 1 |

$12 \div 2 = 6$ remainder 0 |

$6 \div 2 = 3$ remainder 0 |

$3 \div 2 = 1$ remainder 1 |

$1 \div 2 = 0$ remainder 1 | Read upwards

Answer: **110010_2**

Example 4: Convert 7_{10} to base 2

$7 \div 2 = 3$ remainder 1 ↑

3 ÷ 2 = 1 remainder 1 |

1 ÷ 2 = 0 remainder 1 | Read upwards

Answer: **111_2**

Step-by-Step Process Summary

1. Divide the decimal number by 2
2. Record the remainder
3. Use the quotient for the next division
4. Stop when quotient becomes 0
5. Write remainders from bottom to top

EVALUATION

1. Convert 18_{10} to base 2
2. What is 31_{10} in binary?
3. Convert 9_{10} to base 2 and verify your answer
4. Change 40_{10} to base 2
5. Convert 64_{10} to binary

ASSIGNMENT

1. Convert 22_{10} to base 2
2. What is 35_{10} in binary?
3. Convert 100_{10} to base 2
4. Change 17_{10} to binary and verify
5. Convert 63_{10} to base 2

WEEK 8: FRACTIONS I

Learning Objectives

By the end of this lesson, students should be able to:

- Define a fraction
- Identify types of fractions
- Simplify fractions to their lowest terms

CONTENT

WHAT IS A FRACTION?

A fraction represents a part of a whole. It has two parts:

- **Numerator** (top number): shows how many parts we have
- **Denominator** (bottom number): shows how many equal parts the whole is divided into

Example: $\frac{3}{4}$ means 3 parts out of 4 equal parts

TYPES OF FRACTIONS

1. Proper Fractions The numerator is smaller than the denominator. Examples: $\frac{1}{2}$, $\frac{3}{4}$, $\frac{5}{8}$, $\frac{2}{3}$

2. Improper Fractions The numerator is greater than or equal to the denominator. Examples: $\frac{5}{3}$, $\frac{7}{4}$, $\frac{9}{5}$, $\frac{8}{8}$

3. Mixed Numbers A combination of a whole number and a proper fraction. Examples: $1\frac{1}{2}$, $2\frac{3}{4}$, $3\frac{2}{5}$, $5\frac{1}{3}$

CONVERTING BETWEEN IMPROPER FRACTIONS AND MIXED NUMBERS

Improper Fraction to Mixed Number:

Example 1: Convert $\frac{13}{4}$ to a mixed number

- Divide 13 by 4: 13 ÷ 4 = 3 remainder 1
- Whole number = 3, Remainder = 1, Denominator stays 4
- Answer: **$3\frac{1}{4}$**

Example 2: Convert $\frac{23}{5}$ to a mixed number

- 23 ÷ 5 = 4 remainder 3
- Answer: **$4\frac{3}{5}$**

Mixed Number to Improper Fraction:

Example 1: Convert $2\frac{3}{4}$ to an improper fraction

- Multiply whole number by denominator: 2 × 4 = 8
- Add the numerator: 8 + 3 = 11
- Keep the same denominator: 4
- Answer: **$\frac{11}{4}$**

Example 2: Convert $3\frac{2}{5}$ to an improper fraction

- 3 × 5 = 15
- 15 + 2 = 17
- Answer: **$\frac{17}{5}$**

SIMPLIFYING FRACTIONS

To simplify a fraction, divide both the numerator and denominator by their HCF.

Example 1: Simplify $\frac{8}{12}$

Step 1: Find HCF of 8 and 12

- Factors of 8: 1, 2, 4, 8

- Factors of 12: 1, 2, 3, 4, 6, 12
- HCF = 4

Step 2: Divide both by 4

- $\frac{8}{12} = (8 \div 4)/(12 \div 4) = \frac{2}{3}$

Example 2: Simplify $\frac{15}{25}$

- HCF of 15 and 25 = 5
- $\frac{15}{25} = (15 \div 5)/(25 \div 5) = \frac{3}{5}$

Example 3: Simplify $\frac{18}{24}$

- HCF of 18 and 24 = 6
- $\frac{18}{24} = (18 \div 6)/(24 \div 6) = \frac{3}{4}$

EVALUATION

1. Identify whether each fraction is proper, improper, or mixed: $\frac{5}{7}$, $3\frac{1}{2}$, $\frac{9}{4}$
2. Convert $\frac{17}{3}$ to a mixed number
3. Convert $4\frac{2}{5}$ to an improper fraction
4. Simplify $\frac{10}{15}$ to its lowest terms
5. Reduce $\frac{24}{36}$ to its simplest form

ASSIGNMENT

1. Write three examples each of proper and improper fractions
2. Convert $\frac{29}{6}$ to a mixed number
3. Change $5\frac{3}{8}$ to an improper fraction
4. Simplify: (a) $\frac{14}{21}$ (b) $\frac{20}{30}$
5. Express $\frac{36}{48}$ in its lowest terms

WEEK 9: FRACTIONS II

Learning Objectives

By the end of this lesson, students should be able to:

- Find equivalent fractions
- Compare fractions
- Arrange fractions in order

CONTENT

EQUIVALENT FRACTIONS

Equivalent fractions are fractions that have the same value but different numerators and denominators.

Creating Equivalent Fractions: Multiply or divide both numerator and denominator by the same number.

Example 1: Find three equivalent fractions of $\frac{2}{3}$

- $\frac{2}{3} = (2\times2)/(3\times2) = \frac{4}{6}$
- $\frac{2}{3} = (2\times3)/(3\times3) = \frac{6}{9}$
- $\frac{2}{3} = (2\times4)/(3\times4) = \frac{8}{12}$

So, $\frac{2}{3} = \frac{4}{6} = \frac{6}{9} = \frac{8}{12}$

Example 2: Show that $\frac{3}{5}$ and $\frac{12}{20}$ are equivalent

- $\frac{3}{5} = (3\times4)/(5\times4) = \frac{12}{20}$ ✓

Example 3: Find the missing number: $\frac{4}{7}$ = __/21

- 7 × 3 = 21, so multiply numerator by 3
- 4 × 3 = 12

- Answer: **$\frac{12}{21}$**

Comparing Fractions

Method 1: Same Denominator When fractions have the same denominator, the one with the larger numerator is greater.

Example: Compare $\frac{3}{8}$ and $\frac{5}{8}$

- Since 5 > 3, then **$\frac{5}{8} > \frac{3}{8}$**

Method 2: Same Numerator When fractions have the same numerator, the one with the smaller denominator is greater.

Example: Compare $\frac{3}{5}$ and $\frac{3}{7}$

- Since 5 < 7, then **$\frac{3}{5} > \frac{3}{7}$**

Method 3: Different Numerators and Denominators Convert to equivalent fractions with the same denominator (using LCM).

Example 1: Compare $\frac{2}{3}$ and $\frac{3}{4}$

Step 1: Find LCM of 3 and 4 = 12

Step 2: Convert to equivalent fractions

- $\frac{2}{3} = (2\times4)/(3\times4) = \frac{8}{12}$
- $\frac{3}{4} = (3\times3)/(4\times3) = \frac{9}{12}$

Step 3: Compare: 9 > 8

- Therefore, **$\frac{3}{4} > \frac{2}{3}$**

Example 2: Compare $\frac{5}{6}$ and $\frac{7}{9}$

- LCM of 6 and 9 = 18
- $\frac{5}{6} = \frac{15}{18}$
- $\frac{7}{9} = \frac{14}{18}$
- Since 15 > 14, $\frac{5}{6} > \frac{7}{9}$

Arranging Fractions in Order

Example: Arrange in ascending order: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$

Step 1: Find LCM of 3, 4, and 6 = 12

Step 2: Convert all fractions

- $\frac{2}{3} = \frac{8}{12}$
- $\frac{3}{4} = \frac{9}{12}$
- $\frac{5}{6} = \frac{10}{12}$

Step 3: Arrange: 8 < 9 < 10

- Answer: $\frac{2}{3} < \frac{3}{4} < \frac{5}{6}$

EVALUATION

1. Write two equivalent fractions for $\frac{3}{7}$
2. Find the missing number: $\frac{5}{8}$ = 15/__
3. Which is greater: $\frac{4}{9}$ or $\frac{5}{9}$?
4. Compare $\frac{2}{5}$ and $\frac{3}{7}$ using LCM
5. Arrange in descending order: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{5}$

ASSIGNMENT

1. Find three equivalent fractions for $\frac{4}{5}$
2. Complete: $\frac{7}{10}$ = __/30
3. Which is smaller: $\frac{5}{12}$ or $\frac{7}{12}$?
4. Compare $\frac{3}{8}$ and $\frac{5}{12}$
5. Arrange in ascending order: $\frac{2}{5}$, $\frac{3}{10}$, $\frac{1}{2}$, $\frac{3}{4}$

WEEK 10: BASIC OPERATIONS ON WHOLE NUMBERS

Learning Objectives

By the end of this lesson, students should be able to:

- Perform addition of whole numbers
- Subtract whole numbers accurately
- Multiply whole numbers
- Divide whole numbers

ADDITION OF WHOLE NUMBERS

Rules:

1. Arrange numbers in columns according to place value
2. Add from right to left
3. Carry over when necessary

Example 1: 2,456 + 1,783

```
  2,456
+ 1,783
-------
  4,239
```

Example 2: 5,678 + 3,849

```
   1 1 1
  5,678
+ 3,849
-------
  9,527
```

Subtraction of Whole Numbers

Rules:

1. Arrange numbers in columns by place value
2. Subtract from right to left
3. Borrow when necessary

Example 1: 8,567 - 3,425

```
  8,567
- 3,425
-------
  5,142
```

Example 2: 6,035 - 2,748 (with borrowing)

```
  5 9 9 2
  6,0 3 5
- 2,7 4 8
---------
  3,2 8 7
```

Multiplication of Whole Numbers

Example 1: 234 × 5

```
   234
×    5
 -----
 1,170
```

Example 2: 456 × 23

```
    456
  ×  23
  -----
  1,368  (456 × 3)
 9,120   (456 × 20)
 -------
10,488
```

Example 3: 328 × 47

```
    328
  ×  47
  -----
  2,296  (328 × 7)
13,120   (328 × 40)
--------
15,416
```

Division of Whole Numbers

Example 1: 684 ÷ 4

```
     171
   ------
4 | 684
    4↓
    --
    28
    28
    --
     04
      4
     --
      0
```

Answer: **171**

Example 2: 1,456 ÷ 8

```
    182
  -------
8 | 1,456
    8↓
    --
    64
    64
    --
    05
     0
    --
    56
    56
    --
     0
```

Answer: **182**

Example 3: 2,345 ÷ 5

```
    469
  -------
5 | 2,345
    20↓
    ---
    34
    30
    --
    45
    45
    --
     0
```

Answer: **469**

WORD PROBLEMS

Example 1: A school has 1,245 students. If 368 students are in JSS 1, how many students are in other classes?

Solution: 1,245 - 368 = **877 students**

Example 2: If one notebook costs ₦350, how much will 15 notebooks cost?

Solution: 350 × 15 = **₦5,250**

Example 3: A farmer harvested 2,880 oranges. He wants to pack them equally into 12 baskets. How many oranges will be in each basket?

Solution: 2,880 ÷ 12 = **240 oranges**

EVALUATION

1. Add: 4,567 + 2,893
2. Subtract: 8,000 - 3,456
3. Multiply: 345 × 27
4. Divide: 1,845 ÷ 9
5. A book costs ₦1,250. How much will 8 books cost?

ASSIGNMENT

1. Calculate: 6,789 + 3,456 + 1,234
2. Find the difference: 9,500 - 4,768
3. Multiply: 428 × 35
4. Divide: 3,276 ÷ 12

5. A trader bought 24 cartons of drinks. Each carton contains 12 bottles. How many bottles did he buy in total?

REVISION TIPS

Key Topics to Review

1. **Whole Numbers:** Place value, ordering, properties
2. **LCM:** Finding LCM using multiples and prime factorization
3. **HCF:** Finding HCF using factors and prime factorization
4. **Base 2:** Counting in binary, understanding binary numbers
5. **Conversion:** Base 10 to Base 2 using division method
6. **Fractions I:** Types, conversion, simplification
7. **Fractions II:** Equivalent fractions, comparison
8. **Basic Operations:** Addition, subtraction, multiplication, division

GENERAL REVISION QUESTIONS

Whole Numbers & Place Value

1. What is the place value of 7 in 47,685?
2. Arrange in order: 2,456, 987, 3,001, 1,998

LCM and HCF

3. Find the LCM of 8, 12, and 18
4. Calculate the HCF of 24, 36, and 48

Base 2

5. Convert 45_{10} to base 2
6. What is 10110_2 in base 10?

Fractions

7. Simplify $\frac{24}{36}$ to its lowest terms

8. Convert $5\frac{3}{7}$ to an improper fraction
9. Arrange in order: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$
10. Find two equivalent fractions for $\frac{4}{9}$

Basic Operations

11. Calculate: 5,678 + 3,456 - 2,134
12. Multiply: 567 × 34
13. Divide: 4,536 ÷ 18
14. A school has 2,400 students. If $\frac{5}{8}$ of them are boys, how many girls are in the school?
15. If 24 pencils cost ₦1,440, what is the cost of one pencil?

JSS 1 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: ADDITION & SUBTRACTION OF FRACTIONS

Learning Objectives

By the end of this lesson, students should be able to:

- Add fractions with the same denominators
- Add fractions with different denominators
- Subtract fractions with the same denominators
- Subtract fractions with different denominators
- Solve word problems involving addition and subtraction of fractions

ADDITION OF FRACTIONS WITH THE SAME DENOMINATOR

When fractions have the same denominator, simply add the numerators and keep the denominator the same.

Formula: $a/c + b/c = (a + b)/c$

Example 1: Add $^{2}/_{7} + ^{3}/_{7}$

Solution:

- $^{2}/_{7} + ^{3}/_{7} = (2 + 3)/7 = ^{5}/_{7}$

Example 2: Add $^{4}/_{9} + ^{2}/_{9} + ^{1}/_{9}$

Solution:

- $^{4}/_{9} + ^{2}/_{9} + ^{1}/_{9} = (4 + 2 + 1)/9 = ^{7}/_{9}$

Example 3: Add $^{5}/_{12} + ^{7}/_{12}$

Solution:

- $^5/_{12} + ^7/_{12} = (5 + 7)/12 = ^{12}/_{12} =$ **1** (simplify to 1)

ADDITION OF FRACTIONS WITH DIFFERENT DENOMINATORS

When fractions have different denominators, you must first find a common denominator (usually the LCM).

Steps:

1. Find the LCM of the denominators
2. Convert each fraction to an equivalent fraction with the LCM as denominator
3. Add the numerators
4. Simplify if possible

Example 1: Add $^1/_3 + ^1/_4$

Step 1: LCM of 3 and 4 = 12

Step 2: Convert to equivalent fractions

- $^1/_3 = ^4/_{12}$ (multiply by 4/4)
- $^1/_4 = ^3/_{12}$ (multiply by 3/3)

Step 3: Add

- $^4/_{12} + ^3/_{12} = ^7/_{12}$

Answer: $^7/_{12}$

Example 2: Add $^2/_5 + ^3/_{10}$

Step 1: LCM of 5 and 10 = 10

Step 2: Convert

- $^2/_5 = ^4/_{10}$

- $\frac{3}{10} = \frac{3}{10}$

Step 3: Add

- $\frac{4}{10} + \frac{3}{10} = \frac{7}{10}$

Answer: $\mathbf{\frac{7}{10}}$

Example 3: Add $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

Step 1: LCM of 2, 3, and 6 = 6

Step 2: Convert

- $\frac{1}{2} = \frac{3}{6}$
- $\frac{1}{3} = \frac{2}{6}$
- $\frac{1}{6} = \frac{1}{6}$

Step 3: Add

- $\frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{6}{6} = \mathbf{1}$

Example 4: Add $\frac{3}{4} + \frac{5}{6}$

Step 1: LCM of 4 and 6 = 12

Step 2: Convert

- $\frac{3}{4} = \frac{9}{12}$
- $\frac{5}{6} = \frac{10}{12}$

Step 3: Add

- $\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = \mathbf{1\frac{7}{12}}$ (convert to mixed number)

ADDITION OF MIXED NUMBERS

Method 1: Add whole numbers and fractions separately

Example 1: Add $2\frac{3}{5} + 1\frac{2}{5}$

Solution:

- Whole numbers: $2 + 1 = 3$
- Fractions: $\frac{3}{5} + \frac{2}{5} = \frac{5}{5} = 1$
- Total: $3 + 1 =$ **4**

Example 2: Add $3\frac{1}{4} + 2\frac{3}{8}$

Step 1: Add whole numbers: $3 + 2 = 5$

Step 2: Add fractions (LCM of 4 and 8 = 8)

- $\frac{1}{4} = \frac{2}{8}$
- $\frac{2}{8} + \frac{3}{8} = \frac{5}{8}$

Step 3: Combine: $5 + \frac{5}{8} =$ $\mathbf{5\frac{5}{8}}$

Method 2: Convert to improper fractions first

Example: Add $1\frac{3}{4} + 2\frac{1}{2}$

Step 1: Convert to improper fractions

- $1\frac{3}{4} = \frac{7}{4}$
- $2\frac{1}{2} = \frac{5}{2}$

Step 2: Find LCM of 4 and 2 = 4

- $\frac{7}{4} = \frac{7}{4}$
- $\frac{5}{2} = \frac{10}{4}$

Step 3: Add

- $\frac{7}{4} + \frac{10}{4} = \frac{17}{4} =$ $\mathbf{4\frac{1}{4}}$

SUBTRACTION OF FRACTIONS WITH THE SAME DENOMINATOR

When fractions have the same denominator, subtract the numerators and keep the denominator.

Formula: a/c - b/c = (a - b)/c

Example 1: Subtract $^5/_8$ - $^2/_8$

Solution:

- $^5/_8 - {}^2/_8 = (5 - 2)/8 =$ $\mathbf{^3/_8}$

Example 2: Subtract $^7/_9$ - $^4/_9$

Solution:

- $^7/_9 - {}^4/_9 = (7 - 4)/9 =$ $\mathbf{^3/_9 = {}^1/_3}$ (simplified)

SUBTRACTION OF FRACTIONS WITH DIFFERENT DENOMINATORS

Steps:

1. Find the LCM of the denominators
2. Convert to equivalent fractions
3. Subtract the numerators
4. Simplify if possible

Example 1: Subtract $^3/_4$ - $^1/_2$

Step 1: LCM of 4 and 2 = 4

Step 2: Convert

- $^3/_4 = {}^3/_4$
- $^1/_2 = {}^2/_4$

Step 3: Subtract

- $^3/_4 - {}^2/_4 = {}^1/_4$

Answer: $\mathbf{^1/_4}$

Example 2: Subtract $^5/_6$ - $^2/_9$

Step 1: LCM of 6 and 9 = 18

Step 2: Convert

- $^{5}/_{6} = {}^{15}/_{18}$
- $^{2}/_{9} = {}^{4}/_{18}$

Step 3: Subtract

- $^{15}/_{18} - {}^{4}/_{18} = {}^{11}/_{18}$

Answer: $\mathbf{^{11}/_{18}}$

Example 3: Subtract $^{7}/_{10} - {}^{1}/_{5}$

Step 1: LCM of 10 and 5 = 10

Step 2: Convert

- $^{7}/_{10} = {}^{7}/_{10}$
- $^{1}/_{5} = {}^{2}/_{10}$

Step 3: Subtract

- $^{7}/_{10} - {}^{2}/_{10} = {}^{5}/_{10} = \mathbf{^{1}/_{2}}$ (simplified)

SUBTRACTION OF MIXED NUMBERS

Example 1: Subtract $4^{3}/_{5} - 2^{1}/_{5}$

Solution:

- Whole numbers: 4 - 2 = 2
- Fractions: $^{3}/_{5} - {}^{1}/_{5} = {}^{2}/_{5}$
- Answer: $\mathbf{2^{2}/_{5}}$

Example 2: Subtract $5^{2}/_{3} - 2^{3}/_{4}$ (requires borrowing)

Step 1: Convert to improper fractions

- $5^2/_3 = {}^{17}/_3$
- $2^3/_4 = {}^{11}/_4$

Step 2: Find LCM of 3 and 4 = 12

- ${}^{17}/_3 = {}^{68}/_{12}$
- ${}^{11}/_4 = {}^{33}/_{12}$

Step 3: Subtract

- ${}^{68}/_{12} - {}^{33}/_{12} = {}^{35}/_{12} = \mathbf{2^{11}/_{12}}$

Example 3: Subtract $6^1/_4 - 2^3/_4$ (with borrowing)

Step 1: Since ${}^1/_4 < {}^3/_4$, borrow 1 from 6

- $6^1/_4 = 5^5/_4$ ($5 + 1 + {}^1/_4 = 5 + {}^4/_4 + {}^1/_4 = 5^5/_4$)

Step 2: Now subtract

- Whole numbers: 5 - 2 = 3
- Fractions: ${}^5/_4 - {}^3/_4 = {}^2/_4 = {}^1/_2$

Answer: $\mathbf{3^1/_2}$

WORD PROBLEMS

Example 1: Mary ate ${}^1/_4$ of a pizza and John ate ${}^3/_8$ of the same pizza. What fraction of the pizza did they eat together?

Solution:

- ${}^1/_4 + {}^3/_8$
- LCM of 4 and 8 = 8
- ${}^2/_8 + {}^3/_8 = {}^5/_8$

Answer: **They ate ${}^5/_8$ of the pizza**

Example 2: A tank was $^7/_{10}$ full of water. After using $^2/_5$ of the water, what fraction of the tank still has water?

Solution:

- $^7/_{10}$ - $^2/_5$
- LCM of 10 and 5 = 10
- $^7/_{10}$ - $^4/_{10}$ = $^3/_{10}$

Answer: **$^3/_{10}$ of the tank still has water**

Example 3: A farmer planted yam on $^2/_5$ of his land and cassava on $^1/_3$ of his land. What fraction of the land has been planted?

Solution:

- $^2/_5$ + $^1/_3$
- LCM of 5 and 3 = 15
- $^6/_{15}$ + $^5/_{15}$ = $^{11}/_{15}$

Answer: **$^{11}/_{15}$ of the land has been planted**

EVALUATION

1. Add: $^3/_7$ + $^2/_7$
2. Calculate: $^1/_3$ + $^1/_6$
3. Subtract: $^5/_9$ - $^2/_9$
4. Find the difference: $^3/_4$ - $^2/_5$
5. A student spent $^2/_3$ of her money on books and $^1/_6$ on food. What fraction of her money did she spend altogether?

ASSIGNMENT

1. Add: $^{4}/_{11} + {}^{3}/_{11} + {}^{2}/_{11}$
2. Calculate: $^{2}/_{5} + {}^{3}/_{10} + {}^{1}/_{2}$
3. Subtract: $^{7}/_{8} - {}^{3}/_{8}$
4. Find: $5^{3}/_{4} - 2^{1}/_{2}$
5. A rope is $3^{1}/_{2}$ meters long. If $^{13}/_{4}$ meters is cut off, what length of rope remains?

WEEK 2: MULTIPLICATION & DIVISION OF FRACTIONS

Learning Objectives

By the end of this lesson, students should be able to:

- Multiply fractions by whole numbers
- Multiply fractions by fractions
- Divide fractions by whole numbers
- Divide fractions by fractions
- Solve real-life problems involving multiplication and division of fractions

MULTIPLICATION OF FRACTIONS BY WHOLE NUMBERS

To multiply a fraction by a whole number, multiply the numerator by the whole number and keep the denominator.

Formula: $a \times (b/c) = (a \times b)/c$

Example 1: Multiply $3 \times {}^{2}/_{5}$

Solution:

- $3 \times {}^{2}/_{5} = (3 \times 2)/5 = {}^{6}/_{5} = \mathbf{1^{1}/_{5}}$

Example 2: Multiply $4 \times {}^{3}/_{8}$

Solution:

- $4 \times {}^{3}/_{8} = (4 \times 3)/8 = {}^{12}/_{8} = \mathbf{{}^{3}/_{2} = 1^{1}/_{2}}$

Example 3: Multiply $6 \times {}^{5}/_{9}$

Solution:

- $6 \times {}^{5}/_{9} = (6 \times 5)/9 = {}^{30}/_{9} = \mathbf{{}^{10}/_{3} = 3^{1}/_{3}}$

MULTIPLICATION OF FRACTIONS BY FRACTIONS

To multiply fractions, multiply the numerators together and multiply the denominators together.

Formula: $(a/b) \times (c/d) = (a \times c)/(b \times d)$

Example 1: Multiply $\frac{2}{3} \times \frac{3}{5}$

Solution:

- $\frac{2}{3} \times \frac{3}{5} = (2 \times 3)/(3 \times 5) = \frac{6}{15} = \mathbf{\frac{2}{5}}$ (simplified)

Example 2: Multiply $\frac{3}{4} \times \frac{5}{7}$

Solution:

- $\frac{3}{4} \times \frac{5}{7} = (3 \times 5)/(4 \times 7) = \mathbf{\frac{15}{28}}$

Example 3: Multiply $\frac{4}{9} \times \frac{3}{8}$

Solution:

- $\frac{4}{9} \times \frac{3}{8} = (4 \times 3)/(9 \times 8) = \frac{12}{72} = \mathbf{\frac{1}{6}}$ (simplified by dividing by 12)

CANCELLATION METHOD (SIMPLIFICATION BEFORE MULTIPLICATION)

You can simplify before multiplying by cancelling common factors between numerators and denominators.

Example 1: Multiply $\frac{4}{9} \times \frac{3}{8}$

Solution:

- Cancel 4 and 8 (divide both by 4): 4 becomes 1, 8 becomes 2
- Cancel 3 and 9 (divide both by 3): 3 becomes 1, 9 becomes 3
- $\frac{1}{3} \times \frac{1}{2} = \mathbf{\frac{1}{6}}$

Example 2: Multiply $\frac{5}{6} \times \frac{9}{10}$

Solution:

- Cancel 5 and 10 (÷5): 5→1, 10→2
- Cancel 9 and 6 (÷3): 9→3, 6→2
- $^1/_2 \times {}^3/_2 = {}^3/_4$

Answer: **$^3/_4$**

Example 3: Multiply $^8/_{15} \times {}^5/_{12}$

Solution:

- Cancel 8 and 12 (÷4): 8→2, 12→3
- Cancel 5 and 15 (÷5): 5→1, 15→3
- $^2/_3 \times {}^1/_3 = {}^2/_9$

Answer: **$^2/_9$**

MULTIPLICATION OF MIXED NUMBERS

Convert mixed numbers to improper fractions first, then multiply.

Example 1: Multiply $2^1/_3 \times 1^1/_2$

Step 1: Convert to improper fractions

- $2^1/_3 = {}^7/_3$
- $1^1/_2 = {}^3/_2$

Step 2: Multiply

- $^7/_3 \times {}^3/_2 = {}^{21}/_6 = \mathbf{{}^7/_2 = 3^1/_2}$

Example 2: Multiply $3^2/_5 \times 2^1/_4$

Step 1: Convert

- $3^2/_5 = {}^{17}/_5$
- $2^1/_4 = {}^9/_4$

Step 2: Multiply

- $^{17}/_{5} \times {}^{9}/_{4} = {}^{153}/_{20} = \mathbf{7^{13}/_{20}}$

DIVISION OF FRACTIONS BY WHOLE NUMBERS

To divide a fraction by a whole number, multiply the fraction by the reciprocal of the whole number.

Example 1: Divide $^{3}/_{4} \div 2$

Solution:

- $^{3}/_{4} \div 2 = {}^{3}/_{4} \times {}^{1}/_{2} = {}^{3}/_{8}$

Answer: $\mathbf{^{3}/_{8}}$

Example 2: Divide $^{5}/_{6} \div 3$

Solution:

- $^{5}/_{6} \div 3 = {}^{5}/_{6} \times {}^{1}/_{3} = {}^{5}/_{18}$

Answer: $\mathbf{^{5}/_{18}}$

Example 3: Divide $^{7}/_{9} \div 4$

Solution:

- $^{7}/_{9} \div 4 = {}^{7}/_{9} \times {}^{1}/_{4} = {}^{7}/_{36}$

Answer: $\mathbf{^{7}/_{36}}$

DIVISION OF FRACTIONS BY FRACTIONS

To divide by a fraction, multiply by its reciprocal (flip the second fraction).

Formula: (a/b) ÷ (c/d) = (a/b) × (d/c)

Example 1: Divide $^{2}/_{3} \div {}^{4}/_{5}$

Solution:

- $^{2}/_{3} \div {}^{4}/_{5} = {}^{2}/_{3} \times {}^{5}/_{4} = {}^{10}/_{12} = \mathbf{{}^{5}/_{6}}$

Example 2: Divide $^{3}/_{7} \div {}^{2}/_{5}$

Solution:

- $^{3}/_{7} \div {}^{2}/_{5} = {}^{3}/_{7} \times {}^{5}/_{2} = {}^{15}/_{14} = \mathbf{1^{1}/_{14}}$

Example 3: Divide $^{5}/_{8} \div {}^{3}/_{4}$

Solution:

- $^{5}/_{8} \div {}^{3}/_{4} = {}^{5}/_{8} \times {}^{4}/_{3} = {}^{20}/_{24} = \mathbf{{}^{5}/_{6}}$

Example 4: Divide $^{7}/_{10} \div {}^{5}/_{6}$

Solution:

- $^{7}/_{10} \div {}^{5}/_{6} = {}^{7}/_{10} \times {}^{6}/_{5} = {}^{42}/_{50} = \mathbf{{}^{21}/_{25}}$

DIVISION OF MIXED NUMBERS

Convert to improper fractions first, then divide.

Example 1: Divide $2^{1}/_{2} \div 1^{1}/_{4}$

Step 1: Convert

- $2^{1}/_{2} = {}^{5}/_{2}$
- $1^{1}/_{4} = {}^{5}/_{4}$

Step 2: Divide

- $^{5}/_{2} \div {}^{5}/_{4} = {}^{5}/_{2} \times {}^{4}/_{5} = {}^{20}/_{10} = \mathbf{2}$

Example 2: Divide $3^{3}/_{5} \div 1^{4}/_{5}$

Step 1: Convert

- $3^{3}/_{5} = {}^{18}/_{5}$
- $1^{4}/_{5} = {}^{9}/_{5}$

Step 2: Divide

- $^{18}/_5 \div {}^9/_5 = {}^{18}/_5 \times {}^5/_9 = {}^{18}/_9 = \mathbf{2}$

RECIPROCALS

The reciprocal of a fraction is obtained by swapping the numerator and denominator.

Examples:

- Reciprocal of $^3/_4$ is $\mathbf{^4/_3}$
- Reciprocal of $^5/_7$ is $\mathbf{^7/_5}$
- Reciprocal of 2 ($^2/_1$) is $\mathbf{^1/_2}$
- Reciprocal of $1^3/_5$ ($^8/_5$) is $\mathbf{^5/_8}$

REAL-LIFE WORD PROBLEMS

Example 1: A ribbon is $4^1/_2$ meters long. If it is cut into pieces each $^3/_4$ meter long, how many pieces will be obtained?

Solution:

- $4^1/_2 \div {}^3/_4$
- $^9/_2 \div {}^3/_4 = {}^9/_2 \times {}^4/_3 = {}^{36}/_6 =$ **6 pieces**

Example 2: A bottle contains $2^1/_4$ liters of juice. If each glass holds $^3/_8$ liter, how many glasses can be filled?

Solution:

- $2^1/_4 \div {}^3/_8$
- $^9/_4 \div {}^3/_8 = {}^9/_4 \times {}^8/_3 = {}^{72}/_{12} =$ **6 glasses**

Example 3: A bag of rice weighs 12 kg. If $^2/_3$ of it is sold, how many kilograms were sold?

Solution:

- $^2/_3 \times 12 = (2 \times 12)/3 = {}^{24}/_3 =$ **8 kg**

Example 4: A farmer has $3^3/_4$ hectares of land. He wants to divide it equally among his 5 children. How much land will each child get?

Solution:

- $3^3/_4 \div 5$
- $^{15}/_4 \div 5 = {}^{15}/_4 \times {}^1/_5 = {}^{15}/_{20} = {}^3/_4$ **hectare**

Example 5: If $^3/_5$ of a class of 40 students are girls, how many girls are in the class?

Solution:

- $^3/_5 \times 40 = (3 \times 40)/5 = {}^{120}/_5 =$ **24 girls**

EVALUATION

1. Multiply: $^3/_5 \times {}^4/_7$
2. Calculate: $2^1/_3 \times 1^1/_2$
3. Divide: $^5/_6 \div {}^2/_3$
4. Find: $3^1/_4 \div 1^3/_8$
5. A tank can hold 18 liters of water. If it is $^2/_3$ full, how many liters of water are in the tank?

ASSIGNMENT

1. Multiply: $^7/_9 \times {}^3/_{14}$
2. Calculate: $4 \times {}^5/_8$
3. Divide: $^7/_{10} \div 3$
4. Find: $^4/_5 \div {}^2/_3$

5. A piece of cloth $6^{1}/_{2}$ meters long is cut into pieces each $^{11}/_{4}$ meters long. How many pieces are obtained?

WEEK 3: ESTIMATION & APPROXIMATION

Learning Objectives

By the end of this lesson, students should be able to:

- Understand the concept of estimation
- Round off numbers to the nearest 10, 100, and 1000
- Understand significant figures
- Round numbers to a given number of significant figures
- Apply estimation in real-life situations

ESTIMATION

Estimation means finding an approximate value that is close enough to the actual value. We estimate when we don't need an exact answer or when we want to check if our answer is reasonable.

Why do we estimate?

- To check if our calculations are reasonable
- To make quick mental calculations
- To save time when exact answers aren't necessary
- To make predictions

ROUNDING OFF NUMBERS

Rounding off makes numbers simpler to work with while keeping them close to their original value.

ROUNDING TO THE NEAREST 10

Rules:

- Look at the ones digit (last digit)
- If it is 0, 1, 2, 3, or 4 → round down (keep the tens digit the same)
- If it is 5, 6, 7, 8, or 9 → round up (increase the tens digit by 1)
- Replace the ones digit with 0

Examples:

Number	Ones Digit	Rounded to Nearest 10
23	3	20
47	7	50
65	5	70
94	4	90
138	8	140
252	2	250

Example 1: Round 76 to the nearest 10

- Ones digit = 6 (≥5)
- Round up: **80**

Example 2: Round 543 to the nearest 10

- Ones digit = 3 (<5)
- Round down: **540**

ROUNDING TO THE NEAREST 100

Rules:

- Look at the tens digit
- If it is 0, 1, 2, 3, or 4 → round down
- If it is 5, 6, 7, 8, or 9 → round up
- Replace the last two digits with 00

Examples:

Number	Tens Digit	Rounded to Nearest 100
234	3	200
567	6	600
450	5	500
849	4	800
1,378	7	1,400
2,623	2	2,600

Example 1: Round 846 to the nearest 100

- Tens digit = 4 (<5)
- Round down: **800**

Example 2: Round 3,675 to the nearest 100

- Tens digit = 7 (≥5)
- Round up: **3,700**

ROUNDING TO THE NEAREST 1000

Rules:

- Look at the hundreds digit
- If it is 0, 1, 2, 3, or 4 → round down
- If it is 5, 6, 7, 8, or 9 → round up
- Replace the last three digits with 000

Examples:

Number	Hundreds Digit	Rounded to Nearest 1000
2,345	3	2,000
5,678	6	6,000
8,500	5	9,000
12,456	4	12,000
47,823	8	48,000

Example 1: Round 4,367 to the nearest 1000

- Hundreds digit = 3 (<5)
- Round down: **4,000**

Example 2: Round 18,742 to the nearest 1000

- Hundreds digit = 7 (≥5)
- Round up: **19,000**

SIGNIFICANT FIGURES

Significant figures (s.f.) are the meaningful digits in a number that contribute to its precision.

Rules for Identifying Significant Figures:

1. All non-zero digits are significant
   - 234 has **3 significant figures**
   - 5,678 has **4 significant figures**
2. Zeros between non-zero digits are significant
   - 305 has **3 significant figures**
   - 1,002 has **4 significant figures**
3. Leading zeros (zeros before the first non-zero digit) are NOT significant
   - 0.045 has **2 significant figures** (4 and 5)
   - 0.0078 has **2 significant figures** (7 and 8)
4. Trailing zeros (zeros at the end) are significant only if there's a decimal point
   - 450 has **2 significant figures** (4 and 5)
   - 450.0 has **4 significant figures**
   - 0.500 has **3 significant figures**

Examples:

Number	**Number of Significant Figures**	**Significant Digits**
567	3	5, 6, 7
4,050	3	4, 0, 5
0.0023	2	2, 3
300.0	4	3, 0, 0, 0
0.0405	3	4, 0, 5

ROUNDING TO A GIVEN NUMBER OF SIGNIFICANT FIGURES

Steps:

1. Count from the left to find the required number of significant figures
2. Look at the next digit
3. If it's 5 or more, round up; if less than 5, round down

Example 1: Round 4,567 to 2 significant figures

Solution:

- First 2 s.f. are 4 and 5
- Next digit is 6 (≥5)
- Round up: **4,600**

Example 2: Round 0.03845 to 2 significant figures

Solution:

- First 2 s.f. are 3 and 8
- Next digit is 4 (<5)
- Round down: **0.038**

Example 3: Round 23,456 to 3 significant figures

Solution:

- First 3 s.f. are 2, 3, and 4
- Next digit is 5 (≥5)
- Round up: **23,500**

Example 4: Round 0.006789 to 2 significant figures

Solution:

- First 2 s.f. are 6 and 7
- Next digit is 8 (≥5)
- Round up: **0.0068**

Example 5: Round 145,678 to 4 significant figures

Solution:

- First 4 s.f. are 1, 4, 5, and 6
- Next digit is 7 (≥5)
- Round up: **145,700** or **1.457×10^5**

ESTIMATION IN CALCULATIONS

We can estimate answers by rounding numbers first, then calculating.

Example 1: Estimate 48 × 23

Solution:

- Round: 48 ≈ 50, 23 ≈ 20
- 50 × 20 = **1,000**
- (Actual answer: 1,104)

Example 2: Estimate 397 + 612 + 185

Solution:

- Round: 397 ≈ 400, 612 ≈ 600, 185 ≈ 200
- 400 + 600 + 200 = **1,200**
- (Actual answer: 1,194)

Example 3: Estimate 2,456 ÷ 48

Solution:

- Round: 2,456 ≈ 2,400, 48 ≈ 50
- 2,400 ÷ 50 = **48**
- (Actual answer: 51.2)

Example 4: Estimate the cost of 19 books at ₦487 each

Solution:

- Round: 19 ≈ 20, ₦487 ≈ ₦500
- 20 × 500 = **₦10,000**
- (Actual: ₦9,253)

PRACTICAL APPLICATIONS

Example 1: A school has 1,847 students. Estimate to the nearest hundred.

- Answer: **1,800 students**

Example 2: A trader made ₦45,678 in sales. Estimate to 2 significant figures.

- Answer: **₦46,000**

Example 3: The distance between two cities is 287 km. Estimate to the nearest 10.

- Answer: **290 km**

EVALUATION

1. Round 456 to the nearest 10
2. Round 3,478 to the nearest 100
3. How many significant figures are in 0.0450?
4. Round 23,567 to 3 significant figures
5. Estimate the product of 52 × 38 by rounding to the nearest 10

ASSIGNMENT

1. Round 2,847 to the nearest 1000
2. Round 0.06789 to 2 significant figures
3. How many significant figures are in the number 4,050?
4. Round 145,678 to 4 significant figures
5. Estimate the sum of 487 + 623 + 314 by rounding to the nearest 100

WEEK 4: NUMBERS IN BASE 2 – ADDITION

Learning Objectives

By the end of this lesson, students should be able to:

- Understand the rules of binary addition
- Add binary numbers with no carrying
- Add binary numbers with carrying
- Solve multi-digit binary addition problems
- Verify binary addition by converting to base 10

BINARY ADDITION RULES

In binary addition, we only use two digits: 0 and 1. The basic rules are:

+	**0**	**1**
0	0	1
1	1	10

The Four Basic Rules:

1. 0 + 0 = 0
2. 0 + 1 = 1
3. 1 + 0 = 1
4. 1 + 1 = 10 (write 0, carry 1)

Additional Rule with Carrying: 5. 1 + 1 + 1 (when there's a carry) = 11 (write 1, carry 1)

ADDITION WITHOUT CARRYING

Example 1: Add $101_2 + 10_2$

```
  101
+ 10
-------
  111
```

Step-by-step:

- Rightmost: 1 + 0 = 1
- Middle: 0 + 1 = 1
- Leftmost: 1 + 0 = 1

Answer: **111_2**

Verification:

- $101_2 = 5_{10}$
- $10_2 = 2_{10}$
- $111_2 = 7_{10}$
- 5 + 2 = 7 ✓

Example 2: Add $1010_2 + 100_2$

```
 1010
+ 100
--------
 1110
```

Answer: **1110_2**

Verification:

- $1010_2 = 10_{10}$
- $100_2 = 4_{10}$

- $1110_2 = 14_{10}$
- $10 + 4 = 14$ ✓

ADDITION WITH CARRYING

Example 1: Add $11_2 + 1_2$

```
 1
 11
+ 1
-----
100
```

Step-by-step:

- Rightmost: 1 + 1 = 10 (write 0, carry 1)
- Middle: 1 + 0 + 1(carry) = 10 (write 0, carry 1)
- Leftmost: 1(carry)

Answer: **100_2**

Verification:

- $11_2 = 3_{10}$
- $1_2 = 1_{10}$
- $100_2 = 4_{10}$
- $3 + 1 = 4$ ✓

Example 2: Add $111_2 + 11_2$

```
 11
```

```
  111
+  11
-------
 1010
```

Step-by-step:

- Rightmost: 1 + 1 = 10 (write 0, carry 1)
- Middle: 1 + 1 + 1(carry) = 11 (write 1, carry 1)
- Third position: 1 + 0 + 1(carry) = 10 (write 0, carry 1)
- Leftmost: 1(carry)

Answer: **1010_2**

Verification:

- $111_2 = 7_{10}$
- $11_2 = 3_{10}$
- $1010_2 = 10_{10}$
- 7 + 3 = 10 ✓

Example 3: Add $1011_2 + 111_2$

```
  111
 1011
+ 111
--------
10010
```

Step-by-step:

- Position 1: 1 + 1 = 10 (write 0, carry 1)
- Position 2: 1 + 1 + 1 = 11 (write 1, carry 1)
- Position 3: 0 + 1 + 1 = 10 (write 0, carry 1)
- Position 4: 1 + 0 + 1 = 10 (write 0, carry 1)
- Position 5: 1(carry)

Answer: **10010_2**

Verification:

- $1011_2 = 11_{10}$
- $111_2 = 7_{10}$
- $10010_2 = 18_{10}$
- 11 + 7 = 18 ✓

Example 4: Add $1101_2 + 1011_2$

```
 111
 1101
+1011
---------
11000
```

Step-by-step:

- Position 1: 1 + 1 = 10 (write 0, carry 1)
- Position 2: 0 + 1 + 1 = 10 (write 0, carry 1)
- Position 3: 1 + 0 + 1 = 10 (write 0, carry 1)

- Position 4: 1 + 1 + 1 = 11 (write 1, carry 1)
- Position 5: 1(carry)

Answer: **11000_2**

Verification:

- $1101_2 = 13_{10}$
- $1011_2 = 11_{10}$
- $11000_2 = 24_{10}$
- 13 + 11 = 24 ✓

ADDING MORE THAN TWO BINARY NUMBERS

Example 1: Add $101_2 + 11_2 + 110_2$

```
  11
  101
   11
 +110
 -------
 1110
```

Answer: **1110_2**

Verification:

- $101_2 = 5_{10}$
- $11_2 = 3_{10}$
- $110_2 = 6_{10}$
- $1110_2 = 14_{10}$
- 5 + 3 + 6 = 14 ✓

Example 2: Add $111_2 + 101_2 + 11_2$

```
 111
 111
 101
+ 11
-------
10011
```

Answer: **10011_2**

Verification:

- $111_2 = 7_{10}$
- $101_2 = 5_{10}$
- $11_2 = 3_{10}$
- $10011_2 = 19_{10}$
- 7 + 5 + 3 = 15... Wait, let me recalculate.

Actually: 7 + 5 + 3 = 15, and 10011_2 = 16 + 2 + 1 = 19. Let me redo this:

```
 111

 111

 101

+ 11

----

10001
```

Position 1: 1 + 1 + 1 = 11 (write 1, carry 1) Position 2: 1 + 0 + 1 + 1 = 11 (write 1, carry 1)
Wait, let me be more careful:

```
 111  =  7
 101  =  5
```

+ 11 = 3

15

15 in binary is 1111_2

Let me recalculate:

111

111

101

+ 11

1111

Position 1: 1 + 1 + 1 = 11 (write 1, carry 1) Position 2: 1 + 0 + 1 + 1(carry) = 11 (write 1, carry 1) Position 3: 1 + 1 + 0 + 1(carry) = 11 (write 1, carry 1) Position 4: 1(carry)

Answer: **$1111_2 = 15_{10}$ ✓**

WORD PROBLEMS

Example 1: A computer stores 1011_2 bytes in one folder and 110_2 bytes in another. How many bytes are stored in total (in binary)?

Solution:

1011

+110

10001

Answer: **10001_2 bytes**

Example 2: Three binary numbers 100_2, 11_2, and 101_2 are added together. What is the sum?

Solution:

```
  100
   11
 +101
 -------
 1100
```

Answer: **1100_2**

EVALUATION

1. Add: $10_2 + 11_2$
2. Calculate: $101_2 + 100_2$
3. Find the sum: $1101_2 + 111_2$
4. Add: $1001_2 + 1010_2$
5. Calculate and verify: $1111_2 + 1_2$

ASSIGNMENT

1. Add: $110_2 + 101_2$
2. Calculate: $1011_2 + 1001_2$
3. Find the sum: $11_2 + 11_2 + 11_2$
4. Add: $10101_2 + 1010_2$
5. If one file has 1101_2 bytes and another has 1011_2 bytes, what is the total size in binary?

WEEK 6: NUMBERS IN BASE 2 – SUBTRACTION

Learning Objectives

By the end of this lesson, students should be able to:

- Understand the rules of binary subtraction
- Subtract binary numbers without borrowing
- Subtract binary numbers with borrowing
- Solve multi-digit binary subtraction problems
- Verify binary subtraction by converting to base 10

BINARY SUBTRACTION RULES

In binary subtraction, we use these basic rules:

−	**0**	**1**
0	0	Borrow
1	1	0

The Four Basic Rules:

1. 0 - 0 = 0
2. 1 - 0 = 1
3. 1 - 1 = 0
4. 0 - 1 = 1 (but you must borrow 1 from the next higher place value)

When Borrowing:

- $10_2 - 1 = 1$ (since $10_2 = 2_{10}$ in decimal)
- When you borrow in binary, the 1 becomes 10_2 (which equals 2 in decimal)

SUBTRACTION WITHOUT BORROWING

Example 1: Subtract $101_2 - 100_2$

```
  101
 -100
 ------
  001 = 1
```

Step-by-step:

- Rightmost: 1 - 0 = 1
- Middle: 0 - 0 = 0
- Leftmost: 1 - 1 = 0

Answer: $\mathbf{1_2}$

Verification:

- $101_2 = 5_{10}$
- $100_2 = 4_{10}$
- $1_2 = 1_{10}$
- 5 - 4 = 1 ✓

Example 2: Subtract $1101_2 - 1001_2$

```
  1101
 -1001
 -----
  0100 = 100
```

Step-by-step:

- Position 1: 1 - 1 = 0
- Position 2: 0 - 0 = 0
- Position 3: 1 - 0 = 1
- Position 4: 1 - 1 = 0

Answer: **100_2**

Verification:

- $1101_2 = 13_{10}$
- $1001_2 = 9_{10}$
- $100_2 = 4_{10}$
- 13 - 9 = 4 ✓

Example 3: Subtract $111_2 - 101_2$

```
 111
-101
-------
 010 = 10
```

Answer: **10_2**

Verification:

- $111_2 = 7_{10}$
- $101_2 = 5_{10}$
- $10_2 = 2_{10}$
- 7 - 5 = 2 ✓

SUBTRACTION WITH BORROWING

Example 1: Subtract $10_2 - 1_2$

```
  10
- 1
---
  1
```

Step-by-step:

- Rightmost: 0 - 1 requires borrowing
- Borrow 1 from the next position: 10 - 1 = 1
- Leftmost position becomes 0 after lending

Answer: **1_2**

Verification:

- $10_2 = 2_{10}$
- $1_2 = 1_{10}$
- $1_2 = 1_{10}$
- 2 - 1 = 1 ✓

Example 2: Subtract $100_2 - 11_2$

```
0100
  100
- 11
----
   1
```

Step-by-step explanation:

- Position 1: 0 - 1 requires borrowing
- Borrow from position 2, but it's also 0
- Borrow from position 3 (which is 1)
- Position 3: 1 becomes 0
- Position 2: 0 becomes 10, but we need to lend 1, so it becomes 1
- Position 1: 0 becomes 10
- Now: 10 - 1 = 1 in position 1
- Position 2: 1 - 1 = 0

Answer: $\mathbf{1_2}$

Verification:

- $100_2 = 4_{10}$
- $11_2 = 3_{10}$
- $1_2 = 1_{10}$
- 4 - 3 = 1 ✓

Example 3: Subtract $1000_2 - 111_2$

```
 0110
 1000
  -111
 ------
    1
```

Answer: $\mathbf{1_2}$

Verification:

- $1000_2 = 8_{10}$

- $111_2 = 7_{10}$
- $1_2 = 1_{10}$
- 8 - 7 = 1 ✓

Example 4: Subtract $1010_2 - 101_2$

```
10010
1010
-101
------
 101
```

Step-by-step:

- Position 1: 0 - 1 requires borrowing
- Borrow from position 2 (which is 1)
- Position 2: 1 becomes 0
- Position 1: 0 becomes 10, so 10 - 1 = 1
- Position 2: 0 - 0 = 0
- Position 3: 1 - 1 = 0
- Position 4: 1 - 0 = 1

Answer: $\mathbf{101_2}$

Verification:

- $1010_2 = 10_{10}$
- $101_2 = 5_{10}$
- $101_2 = 5_{10}$

- 10 - 5 = 5 ✓

Example 5: Subtract 10101_2 - 1110_2

10101
-1110

0111 = 111

Step-by-step (working from right to left):

- Position 1: 1 - 0 = 1
- Position 2: 0 - 1 requires borrowing
  - Borrow from position 3 (which is 1)
  - Position 3: 1 becomes 0
  - Position 2: 0 becomes 10, so 10 - 1 = 1
- Position 3: 0 - 1 requires borrowing
  - Borrow from position 4 (which is 0)
  - Need to borrow from position 5 (which is 1)
  - Position 5: 1 becomes 0
  - Position 4: 0 becomes 10, lend 1 to position 3, becomes 1
  - Position 3: 0 becomes 10, so 10 - 1 = 1
- Position 4: 1 - 1 = 0
- Position 5: 0 - 0 = 0

Answer: **111_2**

Verification:

- $10101_2 = 21_{10}$
- $1110_2 = 14_{10}$
- $111_2 = 7_{10}$
- 21 - 14 = 7 ✓

STEP-BY-STEP BORROWING METHOD

When you need to borrow:

1. If the next digit to the left is 1, borrow it
   - The 1 becomes 0
   - Your current position becomes 10_2 (=2 in decimal)
2. If the next digit is also 0, keep moving left until you find a 1
   - Change that 1 to 0
   - All the 0s you passed through become 1
   - Your current position becomes 10_2

Example: Subtract 10000_2 - 1_2

```
 10000
 -     1
 ---------
  1111
```

Borrowing process:

- Position 1 is 0, need to borrow
- Positions 2, 3, 4 are all 0
- Borrow from position 5 (which is 1)
- Position 5: 1 → 0

- Position 4: 0 → 1
- Position 3: 0 → 1
- Position 2: 0 → 1
- Position 1: 0 → 10, so 10 - 1 = 1

Answer: **1111_2**

Verification:

- $10000_2 = 16_{10}$
- $1_2 = 1_{10}$
- $1111_2 = 15_{10}$
- 16 - 1 = 15 ✓

WORD PROBLEMS

Example 1: A computer had 1101_2 MB of free space. After installing software, it has 1001_2 MB left. How much space did the software use?

Solution:

```
 1101
 -1001
 --------
  0100 = 100₂
```

Answer: **100_2 MB**

Verification: 13 - 9 = 4, and $100_2 = 4_{10}$ ✓

Example 2: A digital counter shows 11010_2. It decreases by 1101_2. What does it show now?

Solution:

11010

-1101

1101

Answer: **1101_2**

EVALUATION

1. Subtract: $11_2 - 10_2$
2. Calculate: $101_2 - 11_2$
3. Find: $1100_2 - 101_2$
4. Subtract: $10000_2 - 111_2$
5. Calculate and verify: $10101_2 - 1010_2$

ASSIGNMENT

1. Subtract: $110_2 - 100_2$
2. Calculate: $1010_2 - 11_2$
3. Find: $1111_2 - 1010_2$
4. Subtract: $11000_2 - 1011_2$
5. A file was 10110_2 bytes. After compression, it became 1111_2 bytes. How many bytes were saved?

WEEK 8: NUMBERS IN BASE 2 – MULTIPLICATION

Learning Objectives

By the end of this lesson, students should be able to:

- Understand the rules of binary multiplication
- Multiply binary numbers by 0 and 1
- Multiply multi-digit binary numbers
- Use the shift-and-add method
- Verify binary multiplication by converting to base 10

BINARY MULTIPLICATION RULES

Binary multiplication is simpler than decimal multiplication because we only multiply by 0 or 1.

The Basic Rules:

×	**0**	**1**
0	0	0
1	0	1

1. $0 \times 0 = 0$
2. $0 \times 1 = 0$
3. $1 \times 0 = 0$
4. $1 \times 1 = 1$

Key Concept:

- Multiplying by 0 always gives 0
- Multiplying by 1 gives the number itself

SIMPLE BINARY MULTIPLICATION

Example 1: Multiply $101_2 \times 1_2$

```
  101
×  1
------
  101
```

Answer: **101_2**

Verification:

- $101_2 = 5_{10}$
- $1_2 = 1_{10}$
- $5 \times 1 = 5$, and $101_2 = 5_{10}$ ✓

Example 2: Multiply $110_2 \times 10_2$

```
  110
× 10
------
  000  (110 × 0)
 110   (110 × 1, shifted left)
------
 1100
```

Answer: **1100_2**

Verification:

- $110_2 = 6_{10}$

- $10_2 = 2_{10}$
- $1100_2 = 12_{10}$
- $6 \times 2 = 12$ ✓

Example 3: Multiply $11_2 \times 11_2$

```
   11
 × 11
 ------
  11   (11 × 1)
 11    (11 × 1, shifted left)
 -----
 1001
```

Step-by-step addition:

```
  11
 + 110
 --------
 1001
```

Answer: **1001_2**

Verification:

- $11_2 = 3_{10}$
- $3 \times 3 = 9$, and $1001_2 = 9_{10}$ ✓
-

MULTIPLICATION OF LARGER BINARY NUMBERS

Example 1: Multiply $101_2 \times 11_2$

```
  101
×  11
--------
  101  (101 × 1)
 101   (101 × 1, shifted left)
-----
 1111
```

Step-by-step addition:

```
   101
+ 1010
----------
  1111
```

Answer: **1111_2**

Verification:

- $101_2 = 5_{10}$
- $11_2 = 3_{10}$
- $1111_2 = 15_{10}$
- $5 \times 3 = 15$ ✓

Example 2: Multiply $110_2 \times 101_2$

```
   110
×  101
```

------

110 (110 × 1)

000   (110 × 0, shifted left)

110   (110 × 1, shifted left twice)

11110

Step-by-step addition:

110

000

+11000

11110

Answer: **11110_2**

Verification:

- $110_2 = 6_{10}$
- $101_2 = 5_{10}$
- $11110_2 = 30_{10}$
- 6 × 5 = 30 ✓

Example 3: Multiply $111_2 \times 101_2$

111

× 101

----

111 (111 × 1)

000   (111 × 0)

111   (111 × 1, shifted left twice)

100011

Step-by-step addition:

111

000

+ 11100

100011

Answer: **100011_2**

Verification:

- $111_2 = 7_{10}$
- $101_2 = 5_{10}$
- $100011_2 = 35_{10}$
- $7 \times 5 = 35$ ✓

Example 4: Multiply $1011_2 \times 110_2$

1011

× 110

```
------------
    0000  (1011 × 0)
   1011   (1011 × 1, shifted)
  1011    (1011 × 1, shifted twice)
--------
1000010
```

Step-by-step addition:

```
     0000
    10110
 +  101100
----------
  1000010
```

Answer: **1000010_2**

Verification:

- $1011_2 = 11_{10}$
- $110_2 = 6_{10}$
- $1000010_2 = 66_{10}$
- $11 \times 6 = 66$ ✓

THE SHIFT-AND-ADD METHOD

This is the standard method for binary multiplication:

Steps:

1. Write the numbers one under the other
2. For each digit in the multiplier (bottom number):
    - If it's 1, write the multiplicand (top number)
    - If it's 0, write all zeros
3. Shift each row one position to the left
4. Add all the rows together

Example: Multiply $1101_2 \times 1010_2$

```
    1101
 ×   1010
 ---------
    0000  (1101 × 0)
   1101   (1101 × 1, shifted)
  0000    (1101 × 0, shifted twice)
 1101     (1101 × 1, shifted three times)
 ---------
 10000010
```

Answer: $\mathbf{10000010_2}$

Verification:

- $1101_2 = 13_{10}$
- $1010_2 = 10_{10}$
- $10000010_2 = 130_{10}$
- $13 \times 10 = 130$ ✓

SPECIAL CASES

Multiplying by 10_2 (which is 2 in decimal): Multiplying by 10_2 is the same as shifting the number one place to the left (adding a 0 at the end).

Examples:

- $11_2 \times 10_2 = 110_2$ ($3 \times 2 = 6$)
- $101_2 \times 10_2 = 1010_2$ ($5 \times 2 = 10$)
- $111_2 \times 10_2 = 1110_2$ ($7 \times 2 = 14$)

Multiplying by 100_2 (which is 4 in decimal): Shift two places to the left (add two 0s at the end).

Examples:

- $11_2 \times 100_2 = 1100_2$ ($3 \times 4 = 12$)
- $101_2 \times 100_2 = 10100_2$ ($5 \times 4 = 20$)

PRACTICAL PROBLEMS

Example 1: A digital system processes 1011_2 bits per second. How many bits does it process in 11_2 seconds?

Solution:

```
   1011
 ×    11
 -------
   1011
  1011
 -------
```

100001

Answer: **100001_2 bits**

Verification: $11_{10} \times 3_{10} = 33_{10}$, and $100001_2 = 33_{10}$ ✓

Example 2: If one binary packet contains 110_2 bytes and you have 101_2 packets, what is the total size?

Solution:

```
   110
 ×  101
 ------
   110
  000
 110
 ------
 11110
```

Answer: **11110_2 bytes**

Verification: $6_{10} \times 5_{10} = 30_{10}$, and $11110_2 = 30_{10}$ ✓

EVALUATION

1. Multiply: $10_2 \times 11_2$
2. Calculate: $101_2 \times 10_2$
3. Find: $11_2 \times 101_2$
4. Multiply: $110_2 \times 110_2$

5. Calculate and verify: $1001_2 \times 11_2$

ASSIGNMENT

1. Multiply: $111_2 \times 11_2$
2. Calculate: $1010_2 \times 101_2$
3. Find: $1001_2 \times 110_2$
4. Multiply: $1101_2 \times 11_2$
5. A memory chip stores 1011_2 bytes in each sector. If there are 100_2 sectors, what is the total storage in binary?

WEEK 9: ALGEBRAIC PROCESSES I

Learning Objectives

By the end of this lesson, students should be able to:

- Understand what algebra is
- Use letters to represent numbers
- Identify algebraic terms, variables, constants, and coefficients
- Write algebraic expressions
- Translate word problems into algebraic expressions

INTRODUCTION TO ALGEBRA

Algebra is a branch of mathematics that uses letters (called variables) to represent numbers. These letters can stand for unknown values or values that can change.

Why do we use algebra?

- To solve problems with unknown values
- To express general mathematical relationships
- To make calculations shorter and easier
- To represent real-life situations mathematically

ALGEBRAIC SYMBOLS AND TERMS

VARIABLES: A variable is a letter that represents an unknown or changing number. Common variables are: a, b, c, x, y, z, n, m, p, q

Examples:

- Let x represent the number of students

- Let n represent any number
- Let y represent the cost of one book

CONSTANTS: A constant is a fixed number that doesn't change.

Examples:

- In the expression $x + 5$, the number 5 is a constant
- In $3y + 7$, the numbers 3 and 7 are constants

COEFFICIENTS: A coefficient is the number that multiplies the variable.

Examples:

- In $5x$, the coefficient is 5
- In $3y$, the coefficient is 3
- In x (or $1x$), the coefficient is 1
- In $-7n$, the coefficient is -7

TERMS: A term is a single number or variable, or numbers and variables multiplied together.

Examples of terms:

- 5 (constant term)
- x (variable term)
- $3x$ (product of coefficient and variable)
- $7y^2$
- $-4ab$

UNDERSTANDING ALGEBRAIC NOTATION

Multiplication in Algebra: In algebra, we usually don't write the multiplication sign (×).

Examples:

- $3 \times x$ is written as **3x**
- $5 \times y$ is written as **5y**
- $a \times b$ is written as **ab**
- $2 \times m \times n$ is written as **2mn**

Important Notes:

- xy means $x \times y$
- 3ab means $3 \times a \times b$
- The coefficient is always written before the variable
- We write 5x, not x5

Division in Algebra: Division can be written as a fraction.

Examples:

- $x \div 2$ can be written as **x/2** or **½x**
- $a \div b$ can be written as **a/b**
- $(x + 3) \div 4$ can be written as **(x + 3)/4**

WRITING ALGEBRAIC EXPRESSIONS

An algebraic expression is a combination of variables, constants, and operations (+, -, ×, ÷).

Example 1: Write an expression for: "Five more than a number x"

- Answer: **x + 5**

Example 2: Write an expression for: "Three times a number y"

- Answer: **3y**

Example 3: Write an expression for: "A number n decreased by 7"

- Answer: **n - 7**

Example 4: Write an expression for: "The product of 4 and a number m"

- Answer: **4m**

Example 5: Write an expression for: "A number p divided by 6"

- Answer: **p/6** or **p ÷ 6**

Example 6: Write an expression for: "Twice a number plus 8"

- Answer: **2x + 8** (if we let x be the number)

Example 7: Write an expression for: "The sum of three times x and five times y"

- Answer: **3x + 5y**

TRANSLATING WORDS TO ALGEBRA

Addition (+):

- Sum of
- Plus
- More than
- Increased by
- Added to
- Total of

Examples:

- "The sum of x and 7" = **x + 7**
- "5 more than y" = **y + 5**
- "n increased by 3" = **n + 3**

Subtraction (-):

- Difference

- Minus
- Less than
- Decreased by
- Subtracted from
- Fewer than

Examples:

- "x minus 4" = **x - 4**
- "8 less than m" = **m - 8**
- "n decreased by 6" = **n - 6**

Multiplication (×):

- Product of
- Times
- Multiplied by
- Of (when used with fractions)
- Twice, thrice (2×, 3×)

Examples:

- "The product of 6 and y" = **6y**
- "Four times n" = **4n**
- "Twice a number" = **2x**
- "Half of p" = **½p** or **p/2**

Division (÷):

- Quotient of
- Divided by

- Ratio of
- Per

Examples:

- "x divided by 5" = **x/5**
- "The quotient of m and 3" = **m/3**
- "n per unit" = **n/1** or **n**

IDENTIFYING PARTS OF ALGEBRAIC EXPRESSIONS

Example 1: In the expression 7x + 3

- Variable: x
- Coefficient: 7
- Constant: 3
- Terms: 7x and 3 (two terms)

Example 2: In the expression 5y - 8

- Variable: y
- Coefficient: 5
- Constant: -8
- Terms: 5y and -8

Example 3: In the expression 2a + 3b - 4

- Variables: a and b
- Coefficients: 2 and 3
- Constant: -4
- Terms: 2a, 3b, and -4 (three terms)

Example 4: In the expression 9m

- Variable: m
- Coefficient: 9
- Constant: none
- Terms: 9m (one term)

EVALUATING SIMPLE EXPRESSIONS

When you know the value of the variable, you can evaluate (find the value of) the expression.

Example 1: Find the value of $3x + 5$ when $x = 4$

Solution:

- $3x + 5 = 3(4) + 5$
- $= 12 + 5$
- $= \mathbf{17}$

Example 2: Find the value of $2y - 7$ when $y = 10$

Solution:

- $2y - 7 = 2(10) - 7$
- $= 20 - 7$
- $= \mathbf{13}$

Example 3: Find the value of $5n$ when $n = 6$

Solution:

- $5n = 5(6)$
- $= \mathbf{30}$

Example 4: Find the value of $x/2 + 3$ when $x = 8$

Solution:

- $x/2 + 3 = 8/2 + 3$
- $= 4 + 3$
- $= \mathbf{7}$

Example 5: Find the value of $4a + 2b$ when $a = 3$ and $b = 5$

Solution:

- $4a + 2b = 4(3) + 2(5)$
- $= 12 + 10$
- $= \mathbf{22}$

REAL-LIFE APPLICATIONS

Example 1: Let n be the number of notebooks. If each notebook costs ₦50, write an expression for the total cost.

Answer: **50n** or **₦50n**

Example 2: A bus has 45 seats. If x seats are occupied, write an expression for the number of empty seats.

Answer: **45 - x**

Example 3: John has y marbles. His sister has 3 times as many marbles. Write an expression for the number of marbles his sister has.

Answer: **3y**

Example 4: A rectangle has length l and width 5 cm. Write an expression for its perimeter.

Answer: **2l + 10** (since perimeter = 2l + 2w = 2l + 2(5) = 2l + 10)

Example 5: A teacher has n students. She divides them equally into 4 groups. Write an expression for the number of students in each group.

Answer: **n/4**

EVALUATION

1. What is the coefficient in the expression 8x?
2. Write an algebraic expression for "7 less than a number y"
3. In the expression 3m + 5, identify the variable, coefficient, and constant
4. If x = 6, find the value of 4x - 10
5. Write an expression for "the product of 9 and a number n"

ASSIGNMENT

1. Identify the coefficient and constant in: 5y - 12
2. Write an expression for "a number x increased by 15"
3. Translate to algebra: "The quotient of p and 6"
4. Find the value of 2a + 7 when a = 4
5. A pen costs ₦x. Write an expression for the cost of 8 pens.

WEEK 10: SIMPLIFICATION OF ALGEBRAIC EXPRESSIONS

Learning Objectives

By the end of this lesson, students should be able to:

- Identify like and unlike terms
- Collect like terms
- Simplify algebraic expressions by combining like terms
- Expand brackets in algebraic expressions
- Simplify expressions after expanding brackets

LIKE AND UNLIKE TERMS

LIKE TERMS: Like terms are terms that have exactly the same variable(s) raised to the same power.

Examples of like terms:

- $3x$ and $5x$ (both have variable x)
- $7y$ and $-2y$ (both have variable y)
- $4ab$ and $9ab$ (both have variables a and b)
- $2x^2$ and $5x^2$ (both have x^2)

UNLIKE TERMS: Unlike terms have different variables or the same variable raised to different powers.

Examples of unlike terms:

- $3x$ and $5y$ (different variables)
- $4x$ and $4x^2$ (different powers)

- 2ab and 2ac (different second variable)
- 5 and 5x (one is a constant, one has a variable)

More Examples:

Which terms are like terms?

- 2x, 5x, -3x are **like terms**
- 4y, 7y are **like terms**
- 3x, 3y are **unlike terms**
- 5ab, 2ab, -ab are **like terms**
- $3x^2$, 5x are **unlike terms**

COLLECTING LIKE TERMS

To simplify expressions, we combine (add or subtract) like terms.

Rules:

1. Only like terms can be combined
2. Add or subtract the coefficients
3. Keep the variable part the same

Example 1: Simplify 3x + 5x

Solution:

- Both terms have x
- Add coefficients: 3 + 5 = 8
- Answer: **8x**

Example 2: Simplify 7y - 2y

Solution:

- Both terms have y
- Subtract coefficients: 7 - 2 = 5
- Answer: **5y**

Example 3: Simplify 4a + 3a + 2a

Solution:

- All terms have a
- Add coefficients: 4 + 3 + 2 = 9
- Answer: **9a**

Example 4: Simplify 8m - 5m + 3m

Solution:

- All terms have m
- Calculate: 8 - 5 + 3 = 6
- Answer: **6m**

Example 5: Simplify 9x - 12x

Solution:

- Both terms have x
- Subtract: 9 - 12 = -3
- Answer: **-3x**

SIMPLIFYING EXPRESSIONS WITH MULTIPLE VARIABLES

Example 1: Simplify 5x + 3y + 2x + 4y

Solution:

- Collect like terms:

- x terms: 5x + 2x = 7x
- y terms: 3y + 4y = 7y

- Answer: **7x + 7y**

Example 2: Simplify 4a + 3b - 2a + 5b

Solution:

- Collect like terms:
  - a terms: 4a - 2a = 2a
  - b terms: 3b + 5b = 8b
- Answer: **2a + 8b**

Example 3: Simplify 7m + 5n - 3m - 2n

Solution:

- m terms: 7m - 3m = 4m
- n terms: 5n - 2n = 3n
- Answer: **4m + 3n**

Example 4: Simplify 6p + 4q + 2r - 2p - q + 3r

Solution:

- p terms: 6p - 2p = 4p
- q terms: 4q - q = 3q
- r terms: 2r + 3r = 5r
- Answer: **4p + 3q + 5r**

SIMPLIFYING EXPRESSIONS WITH CONSTANTS

Example 1: Simplify 3x + 5 + 2x + 7

Solution:

- x terms: 3x + 2x = 5x
- Constants: 5 + 7 = 12
- Answer: **5x + 12**

Example 2: Simplify 4y + 8 - 2y - 3

Solution:

- y terms: 4y - 2y = 2y
- Constants: 8 - 3 = 5
- Answer: **2y + 5**

Example 3: Simplify 5a + 3b + 7 - 2a + 4b - 2

Solution:

- a terms: 5a - 2a = 3a
- b terms: 3b + 4b = 7b
- Constants: 7 - 2 = 5
- Answer: **3a + 7b + 5**

Example 4: Simplify 9x - 6 + 4x + 10 - 2x

Solution:

- x terms: 9x + 4x - 2x = 11x
- Constants: -6 + 10 = 4
- Answer: **11x + 4**

EXPANDING BRACKETS

When we multiply a number or variable by an expression in brackets, we multiply each term inside the bracket by the number outside.

Formula: $a(b + c) = ab + ac$

This is called the **DISTRIBUTIVE LAW**.

Example 1: Expand $3(x + 4)$

Solution:

- Multiply each term by 3
- $3 \times x = 3x$
- $3 \times 4 = 12$
- Answer: $\mathbf{3x + 12}$

Example 2: Expand $5(y - 3)$

Solution:

- $5 \times y = 5y$
- $5 \times (-3) = -15$
- Answer: $\mathbf{5y - 15}$

Example 3: Expand $2(3x + 5)$

Solution:

- $2 \times 3x = 6x$
- $2 \times 5 = 10$
- Answer: $\mathbf{6x + 10}$

Example 4: Expand $4(2a - 7)$

Solution:

- $4 \times 2a = 8a$

- $4 \times (-7) = -28$
- Answer: **8a - 28**

Example 5: Expand -3(x + 5)

Solution:

- $-3 \times x = -3x$
- $-3 \times 5 = -15$
- Answer: **-3x - 15**

Example 6: Expand 6(2m - 3n + 4)

Solution:

- $6 \times 2m = 12m$
- $6 \times (-3n) = -18n$
- $6 \times 4 = 24$
- Answer: **12m - 18n + 24**

EXPANDING AND SIMPLIFYING

Sometimes we need to expand brackets and then collect like terms.

Example 1: Simplify 2(x + 3) + 4(x + 1)

Solution: Step 1: Expand each bracket

- $2(x + 3) = 2x + 6$
- $4(x + 1) = 4x + 4$

Step 2: Collect like terms

- $2x + 6 + 4x + 4$
- x terms: $2x + 4x = 6x$

- Constants: 6 + 4 = 10

Answer: **6x + 10**

Example 2: Simplify 3(2y + 5) - 2(y - 3)

Solution: Step 1: Expand

- 3(2y + 5) = 6y + 15
- -2(y - 3) = -2y + 6

Step 2: Collect like terms

- 6y + 15 - 2y + 6
- y terms: 6y - 2y = 4y
- Constants: 15 + 6 = 21

Answer: **4y + 21**

Example 3: Simplify 5(a + 2) + 3(a - 4) - 2(a + 1)

Solution: Step 1: Expand all brackets

- 5(a + 2) = 5a + 10
- 3(a - 4) = 3a - 12
- -2(a + 1) = -2a - 2

Step 2: Collect like terms

- 5a + 10 + 3a - 12 - 2a - 2
- a terms: 5a + 3a - 2a = 6a
- Constants: 10 - 12 - 2 = -4

Answer: **6a - 4**

Example 4: Simplify 4(2x + 3y) - 3(x - 2y)

Solution: Step 1: Expand

- $4(2x + 3y) = 8x + 12y$
- $-3(x - 2y) = -3x + 6y$

Step 2: Collect like terms

- $8x + 12y - 3x + 6y$
- x terms: $8x - 3x = 5x$
- y terms: $12y + 6y = 18y$

Answer: **$5x + 18y$**

REMOVING BRACKETS WITH VARIABLES

Example 1: Expand $x(y + 5)$

Solution:

- $x \times y = xy$
- $x \times 5 = 5x$
- Answer: **$xy + 5x$**

Example 2: Expand $a(b - 3)$

Solution:

- $a \times b = ab$
- $a \times (-3) = -3a$
- Answer: **$ab - 3a$**

Example 3: Expand $2m(n + 4)$

Solution:

- $2m \times n = 2mn$
- $2m \times 4 = 8m$

- Answer: **2mn + 8m**

Example 4: Expand $3x(2x + 5)$

Solution:

- $3x \times 2x = 6x^2$
- $3x \times 5 = 15x$
- Answer: $\mathbf{6x^2 + 15x}$

COMMON ERRORS TO AVOID

1. **Don't add unlike terms**
    - Wrong: $3x + 2y = 5xy$
    - Right: $3x + 2y$ (cannot be simplified further)
2. **Don't forget to multiply ALL terms in the bracket**
    - Wrong: $3(x + 2) = 3x + 2$
    - Right: $3(x + 2) = 3x + 6$
3. **Be careful with negative signs**
    - Wrong: $-(x + 3) = -x + 3$
    - Right: $-(x + 3) = -x - 3$
4. **Remember the coefficient**
    - Wrong: $x + x = x$
    - Right: $x + x = 2x$

EVALUATION

1. Simplify: $7x + 3x - 2x$

2. Collect like terms: 5a + 3b - 2a + 4b
3. Expand: 4(x + 6)
4. Simplify: 3(y + 2) + 2(y - 1)
5. Expand and simplify: 5(2m - 3) - 2(m + 4)

ASSIGNMENT

1. Simplify: 9y - 4y + 6y
2. Collect like terms: 7p + 4q - 3p - 2q + 5
3. Expand: 6(3n - 5)
4. Simplify: 2(3x + 4) + 5(x - 2)
5. Expand and simplify: 4(2a + 3b) - 3(a - 2b)

REVISION TIPS (WEEKS 11-13)

Key Topics to Review

1. **Fractions:** Addition, subtraction, multiplication, and division of fractions
2. **Estimation:** Rounding to nearest 10, 100, 1000, and significant figures
3. **Base 2 Addition:** Binary addition with and without carrying
4. **Base 2 Subtraction:** Binary subtraction with and without borrowing
5. **Base 2 Multiplication:** Multiplying binary numbers
6. **Algebraic Processes:** Variables, coefficients, constants, writing expressions
7. **Simplification:** Collecting like terms, expanding brackets

Study Strategies

- Practice all worked examples again
- Complete all evaluation and assignment questions
- Form study groups and explain concepts to each other
- Create summary notes for each topic
- Practice converting between base 2 and base 10
- Master the rules of binary operations
- Solve word problems daily
- Review your mistakes and understand why they were wrong

COMPREHENSIVE REVISION QUESTIONS

Fractions

1. Calculate: $\frac{3}{5} + \frac{2}{3}$
2. Find: $2\frac{1}{4} - 1\frac{3}{8}$

3. Multiply: $^5/_6 \times ^4/_{15}$

4. Divide: $3^3/_5 \div 1^1/_5$

5. A rope $5^1/_2$ m long is cut into pieces each $^3/_4$ m. How many pieces?

Estimation & Approximation

6. Round 4,567 to the nearest hundred
7. Round 0.04569 to 2 significant figures
8. Estimate: 48 × 52 (round to nearest 10)
9. How many s.f. are in 0.00450?
10. Round 234,567 to 3 significant figures

Base 2 Operations

11. Add: $1101_2 + 1011_2$
12. Subtract: $10110_2 - 1101_2$
13. Multiply: $110_2 \times 101_2$
14. Convert 45_{10} to base 2 and add it to 10111_2
15. Verify: $1111_2 - 1010_2$ by converting to base 10

Algebra

16. Write an expression: "8 more than twice a number x"
17. If y = 5, find the value of 3y - 7
18. Identify the coefficient in: 9m - 4
19. Expand: 5(3x - 4)
20. Simplify: 4(2a + 3) - 3(a - 2)

JSS 1 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: SIMPLE EQUATIONS

Learning Objectives

By the end of this lesson, students should be able to:

- Understand what an equation is
- Identify the parts of an equation
- Solve simple one-variable linear equations
- Check solutions to equations
- Apply equations to solve real-life problems

WHAT IS AN EQUATION?

An equation is a mathematical statement that shows two expressions are equal. It always contains an equals sign (=).

Examples of equations:

- $x + 5 = 12$
- $2y - 3 = 7$
- $3n = 15$
- $m/4 = 8$

Parts of an equation:

- **Left-hand side (LHS):** The expression before the equals sign
- **Right-hand side (RHS):** The expression after the equals sign
- **Variable:** The unknown value (usually a letter)

- **Solution:** The value of the variable that makes the equation true

Example: In the equation x + 5 = 12

- LHS = x + 5
- RHS = 12
- Variable = x
- Solution = 7 (because 7 + 5 = 12)

DIFFERENCE BETWEEN EXPRESSIONS AND EQUATIONS

Expression: A mathematical phrase without an equals sign

- Examples: 3x + 5, 2y - 7, 4m

Equation: A mathematical statement with an equals sign

- Examples: 3x + 5 = 14, 2y - 7 = 3, 4m = 20

SOLVING SIMPLE EQUATIONS

To solve an equation means to find the value of the variable that makes the equation true.

SOLVING EQUATIONS BY ADDITION

When a number is subtracted from the variable, add the same number to both sides.

Example 1: Solve x - 5 = 8

Solution:

- x - 5 = 8
- Add 5 to both sides: x - 5 + 5 = 8 + 5
- x = 13

Check: 13 - 5 = 8 ✓

Example 2: Solve y - 12 = 20

Solution:

- y - 12 = 20
- Add 12 to both sides: y - 12 + 12 = 20 + 12
- y = 32

Check: 32 - 12 = 20 ✓

Example 3: Solve m - 7 = 15

Solution:

- m - 7 = 15
- m - 7 + 7 = 15 + 7
- m = 22

Check: 22 - 7 = 15 ✓

SOLVING EQUATIONS BY SUBTRACTION

When a number is added to the variable, subtract the same number from both sides.

Example 1: Solve x + 6 = 14

Solution:

- x + 6 = 14
- Subtract 6 from both sides: x + 6 - 6 = 14 - 6
- x = 8

Check: 8 + 6 = 14 ✓

Example 2: Solve n + 15 = 23

Solution:

- n + 15 = 23
- n + 15 - 15 = 23 - 15
- n = 8

Check: 8 + 15 = 23 ✓

Example 3: Solve p + 9 = 25

Solution:

- p + 9 = 25
- p + 9 - 9 = 25 - 9
- p = 16

Check: 16 + 9 = 25 ✓

SOLVING EQUATIONS BY MULTIPLICATION

When the variable is divided by a number, multiply both sides by that number.

Example 1: Solve x/3 = 5

Solution:

- x/3 = 5
- Multiply both sides by 3: (x/3) × 3 = 5 × 3
- x = 15

Check: 15/3 = 5 ✓

Example 2: Solve y/4 = 7

Solution:

- y/4 = 7
- Multiply both sides by 4: y = 7 × 4
- y = 28

Check: 28/4 = 7 ✓

Example 3: Solve m/5 = 9

Solution:

- m/5 = 9
- m = 9 × 5
- m = 45

Check: 45/5 = 9 ✓

SOLVING EQUATIONS BY DIVISION

When the variable is multiplied by a number, divide both sides by that number.

Example 1: Solve 4x = 20

Solution:

- 4x = 20
- Divide both sides by 4: 4x/4 = 20/4
- x = 5

Check: 4 × 5 = 20 ✓

Example 2: Solve 7y = 42

Solution:

- 7y = 42
- y = 42/7
- y = 6

Check: 7 × 6 = 42 ✓

Example 3: Solve 9n = 63

Solution:

- 9n = 63
- n = 63/9
- n = 7

Check: 9 × 7 = 63 ✓

Example 4: Solve 12m = 84

Solution:

- 12m = 84
- m = 84/12
- m = 7

Check: 12 × 7 = 84 ✓

SOLVING TWO-STEP EQUATIONS

Some equations require more than one operation to solve.

Example 1: Solve 2x + 5 = 13

Solution: Step 1: Subtract 5 from both sides

- 2x + 5 - 5 = 13 - 5

- $2x = 8$

Step 2: Divide both sides by 2

- $2x/2 = 8/2$
- $x = 4$

Check: $2(4) + 5 = 8 + 5 = 13$ ✓

Example 2: Solve $3y - 7 = 14$

Solution: Step 1: Add 7 to both sides

- $3y - 7 + 7 = 14 + 7$
- $3y = 21$

Step 2: Divide both sides by 3

- $y = 21/3$
- $y = 7$

Check: $3(7) - 7 = 21 - 7 = 14$ ✓

Example 3: Solve $5m + 8 = 33$

Solution: Step 1: Subtract 8

- $5m = 33 - 8$
- $5m = 25$

Step 2: Divide by 5

- $m = 25/5$
- $m = 5$

Check: $5(5) + 8 = 25 + 8 = 33$ ✓

Example 4: Solve $4n - 12 = 20$

Solution: Step 1: Add 12

- 4n = 20 + 12
- 4n = 32

Step 2: Divide by 4

- n = 32/4
- n = 8

Check: 4(8) - 12 = 32 - 12 = 20 ✓

Example 5: Solve x/2 + 3 = 10

Solution: Step 1: Subtract 3

- x/2 = 10 - 3
- x/2 = 7

Step 2: Multiply by 2

- x = 7 × 2
- x = 14

Check: 14/2 + 3 = 7 + 3 = 10 ✓

EQUATIONS WITH VARIABLES ON BOTH SIDES

Example 1: Solve 5x = 3x + 10

Solution: Step 1: Subtract 3x from both sides

- 5x - 3x = 3x + 10 - 3x
- 2x = 10

Step 2: Divide by 2

- x = 10/2
- x = 5

Check: 5(5) = 3(5) + 10 → 25 = 15 + 10 → 25 = 25 ✓

Example 2: Solve 7y - 4 = 5y + 6

Solution: Step 1: Subtract 5y from both sides

- 7y - 5y - 4 = 5y - 5y + 6
- 2y - 4 = 6

Step 2: Add 4 to both sides

- 2y = 6 + 4
- 2y = 10

Step 3: Divide by 2

- y = 5

Check: 7(5) - 4 = 5(5) + 6 → 35 - 4 = 25 + 6 → 31 = 31 ✓

WORD PROBLEMS INVOLVING EQUATIONS

Example 1: A number increased by 7 gives 20. Find the number.

Solution: Let the number be x

- x + 7 = 20
- x = 20 - 7
- x = 13

Answer: **The number is 13**

Example 2: When 5 is subtracted from a number, the result is 18. What is the number?

Solution: Let the number be n

- $n - 5 = 18$
- $n = 18 + 5$
- $n = 23$

Answer: **The number is 23**

Example 3: Three times a number is 42. Find the number.

Solution: Let the number be x

- $3x = 42$
- $x = 42/3$
- $x = 14$

Answer: **The number is 14**

Example 4: A number divided by 4 equals 9. Find the number.

Solution: Let the number be y

- $y/4 = 9$
- $y = 9 \times 4$
- $y = 36$

Answer: **The number is 36**

Example 5: John is x years old. In 5 years, he will be 18 years old. How old is he now?

Solution:

- $x + 5 = 18$
- $x = 18 - 5$
- $x = 13$

Answer: **John is 13 years old now**

Example 6: A pen costs ₦x. Five pens cost ₦75. How much is one pen?

Solution:

- $5x = 75$
- $x = 75/5$
- $x = 15$

Answer: **One pen costs ₦15**

Example 7: Mary has some money. After spending ₦250, she has ₦150 left. How much did she have at first?

Solution: Let the original amount be m

- $m - 250 = 150$
- $m = 150 + 250$
- $m = 400$

Answer: **She had ₦400 at first**

IMPORTANT RULES FOR SOLVING EQUATIONS

1. **Balance Rule:** Whatever you do to one side, do to the other side
2. **Inverse Operations:** Use opposite operations to isolate the variable
   - Addition ↔ Subtraction
   - Multiplication ↔ Division
3. **Always check your answer** by substituting back into the original equation
4. **Simplify first** if there are like terms on either side

EVALUATION

1. Solve: $x + 9 = 21$
2. Find the value of y if $4y = 36$
3. Solve: $m - 8 = 15$
4. Calculate: $3n + 5 = 20$
5. A number multiplied by 6 gives 48. What is the number?

ASSIGNMENT

1. Solve: $p + 12 = 30$
2. Find x if $x/5 = 7$
3. Solve: $2y - 3 = 11$
4. Calculate: $5m + 8 = 33$
5. When 7 is added to a number, the result is 25. Find the number.

WEEK 2: PLANE SHAPES

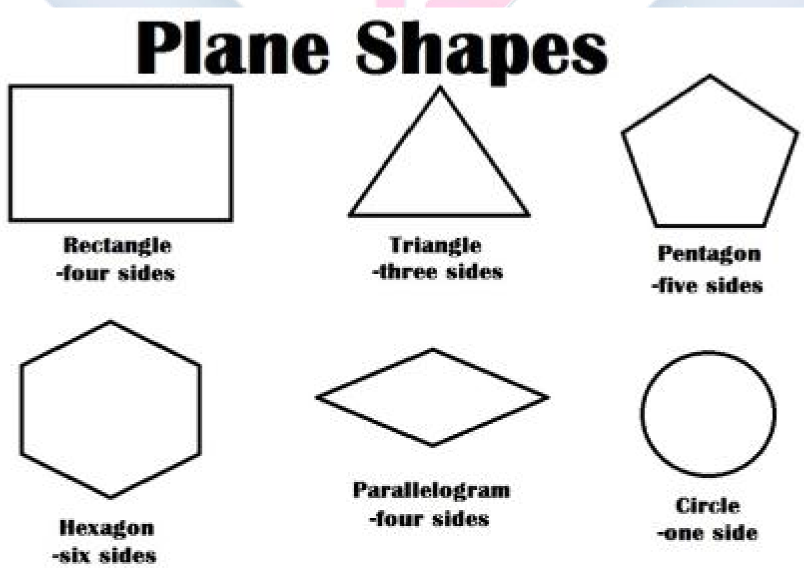
Learning Objectives

By the end of this lesson, students should be able to:

- Identify different plane shapes
- State the properties of triangles
- State the properties of quadrilaterals
- Identify and describe circles
- Calculate perimeters and areas of plane shapes

WHAT ARE PLANE SHAPES?

Plane shapes (or 2D shapes) are flat shapes that have only two dimensions: length and width. They have no thickness.



Common plane shapes:

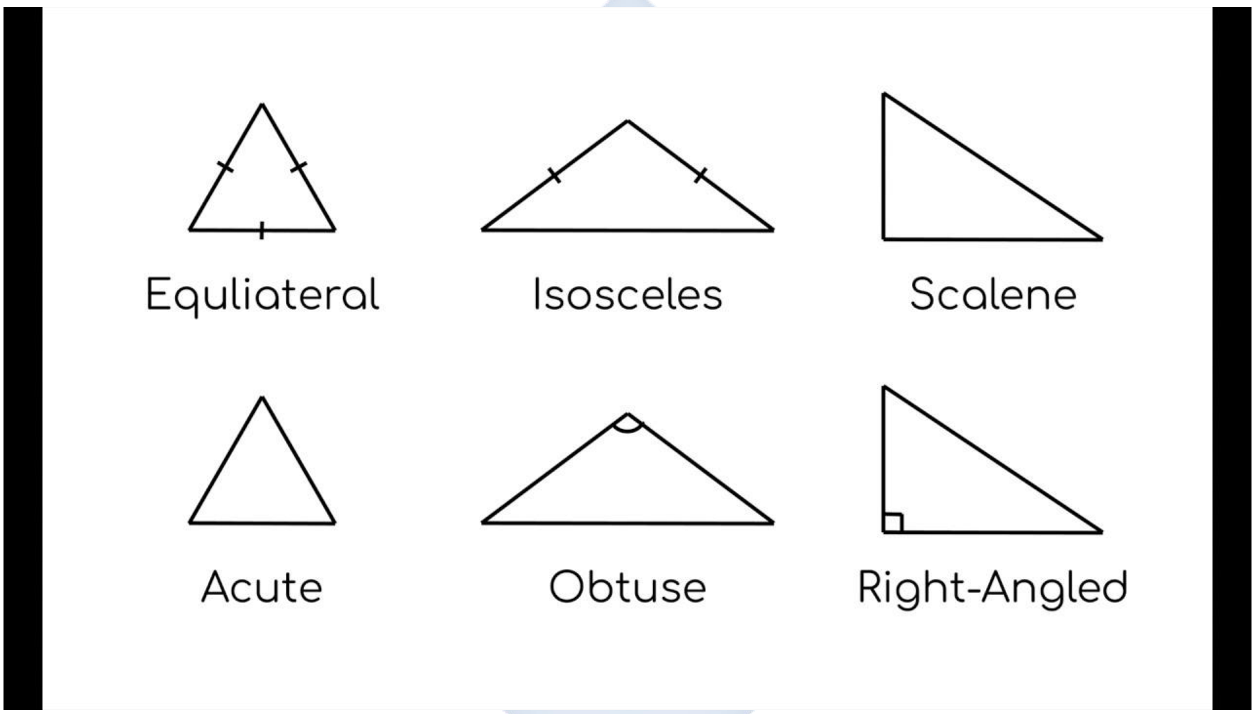
- Triangles
- Quadrilaterals (squares, rectangles, parallelograms, rhombus, trapezium)

- Circles
- Polygons

TRIANGLES

A triangle is a plane shape with three sides and three angles.

TYPES OF TRIANGLES (BY SIDES)



1. EQUILATERAL TRIANGLE

- All three sides are equal in length
- All three angles are equal (60° each)
- Has three lines of symmetry

Properties:

- AB = BC = AC
- ∠A = ∠B = ∠C = 60°

2. ISOSCELES TRIANGLE

- Two sides are equal in length
- Two angles are equal (the angles opposite the equal sides)
- Has one line of symmetry

Properties:

- If $AB = AC$, then $\angle B = \angle C$

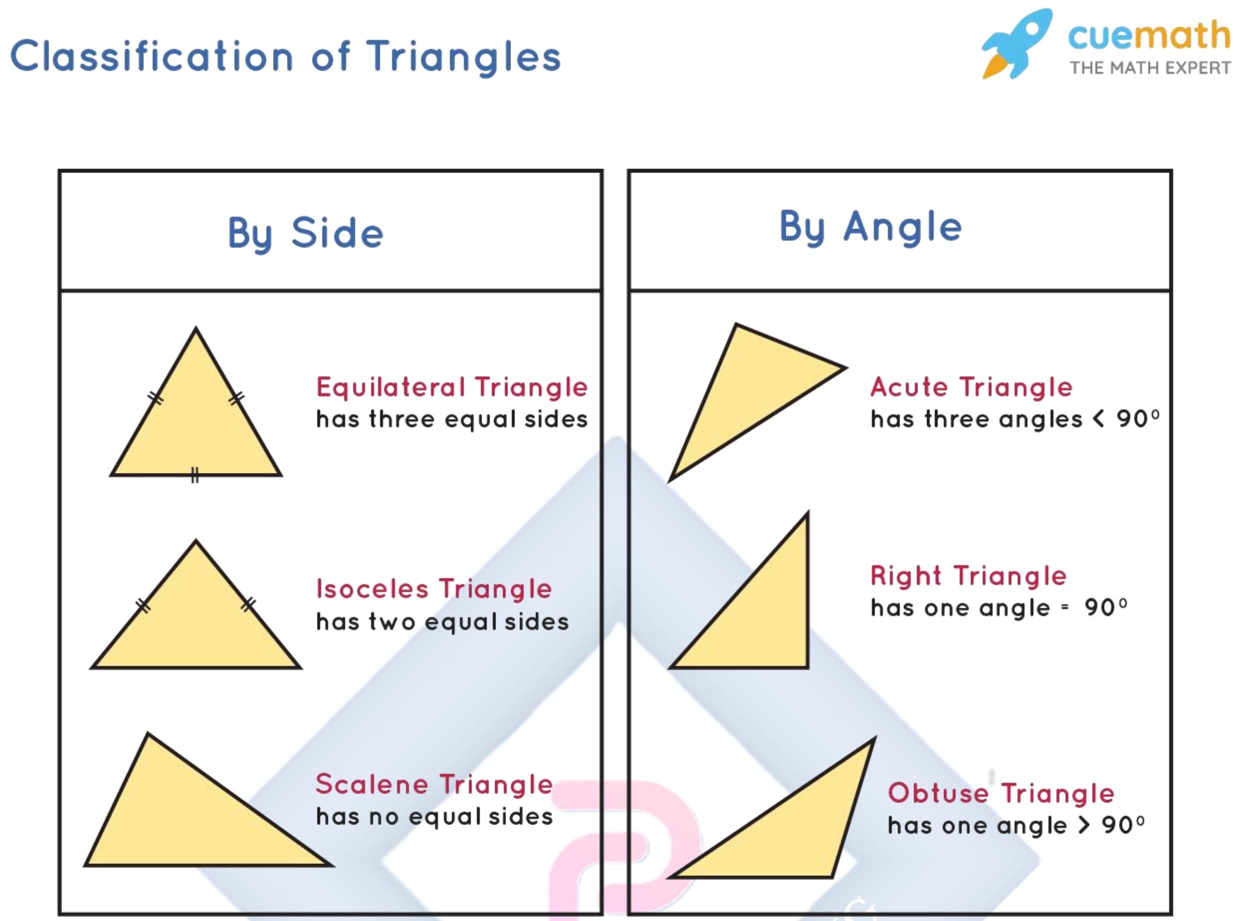
3. SCALENE TRIANGLE

- All three sides have different lengths
- All three angles are different
- No lines of symmetry

Properties:

- $AB \neq BC \neq AC$
- $\angle A \neq \angle B \neq \angle C$

TYPES OF TRIANGLES (BY ANGLES)



1. ACUTE-ANGLED TRIANGLE

- All three angles are less than 90°
- All angles are acute angles

2. RIGHT-ANGLED TRIANGLE

- One angle is exactly 90° (a right angle)
- The other two angles are acute and add up to 90°
- The side opposite the right angle is called the hypotenuse

3. OBTUSE-ANGLED TRIANGLE

- One angle is greater than 90° (an obtuse angle)
- The other two angles are acute

PROPERTIES OF TRIANGLES

1. **Sum of angles:** The sum of all three angles in any triangle is 180°
   - ∠A + ∠B + ∠C = 180°
2. **Sum of any two sides:** The sum of any two sides of a triangle is greater than the third side
   - AB + BC > AC
3. **Perimeter:** Sum of all three sides
   - Perimeter = a + b + c
4. **Area of a triangle:**
   - Area = ½ × base × height

Example 1: Find the third angle of a triangle if two angles are 50° and 70°.

Solution:

- Sum of angles = 180°
- 50° + 70° + third angle = 180°
- 120° + third angle = 180°
- Third angle = 180° - 120° = **60°**

Example 2: The sides of a triangle are 5 cm, 7 cm, and 9 cm. Find its perimeter.

Solution:

- Perimeter = 5 + 7 + 9 = **21 cm**

Example 3: A triangle has a base of 8 cm and height of 6 cm. Find its area.

Solution:

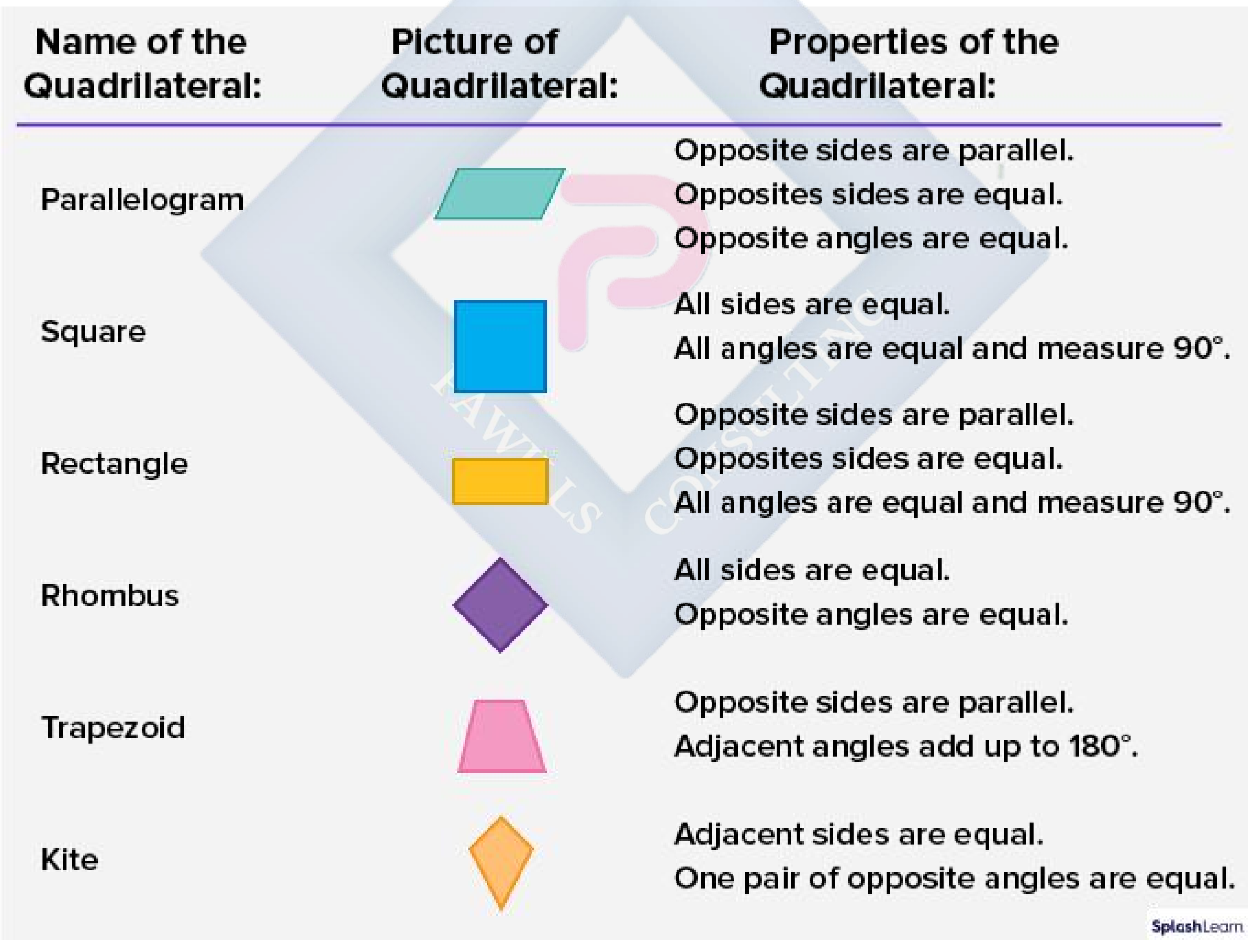
- Area = ½ × base × height

- Area = ½ × 8 × 6
- Area = ½ × 48
- Area = **24 cm²**

QUADRILATERALS

A quadrilateral is a plane shape with four sides and four angles.

TYPES OF QUADRILATERALS



Name of the Quadrilateral:	Picture of Quadrilateral:	Properties of the Quadrilateral:
Parallelogram		Opposite sides are parallel. Opposites sides are equal. Opposite angles are equal.
Square		All sides are equal. All angles are equal and measure 90°.
Rectangle		Opposite sides are parallel. Opposites sides are equal. All angles are equal and measure 90°.
Rhombus		All sides are equal. Opposite angles are equal.
Trapezoid		Opposite sides are parallel. Adjacent angles add up to 180°.
Kite		Adjacent sides are equal. One pair of opposite angles are equal.

SplashLearn

1. SQUARE

Properties:

- All four sides are equal
- All four angles are 90° (right angles)
- Opposite sides are parallel
- Diagonals are equal and bisect each other at right angles
- Has 4 lines of symmetry

Formulas:

- Perimeter = 4 × side = 4s
- Area = side × side = s^2

Example: A square has sides of 7 cm. Find its perimeter and area.

- Perimeter = 4 × 7 = **28 cm**
- Area = 7 × 7 = **49 cm²**

2. RECTANGLE

Properties:

- Opposite sides are equal and parallel
- All four angles are 90°
- Diagonals are equal and bisect each other
- Has 2 lines of symmetry

Formulas:

- Perimeter = 2(length + width) = 2(l + w)
- Area = length × width = l × w

Example: A rectangle has length 12 cm and width 5 cm. Find its perimeter and area.

- Perimeter = 2(12 + 5) = 2(17) = **34 cm**
- Area = 12 × 5 = **60 cm²**

3. PARALLELOGRAM

Properties:

- Opposite sides are equal and parallel
- Opposite angles are equal
- Adjacent angles are supplementary (add up to 180°)
- Diagonals bisect each other

Formulas:

- Perimeter = 2(a + b) where a and b are adjacent sides
- Area = base × height

Example: A parallelogram has base 10 cm and height 6 cm. Find its area.

- Area = 10 × 6 = **60 cm²**

4. RHOMBUS

Properties:

- All four sides are equal
- Opposite sides are parallel
- Opposite angles are equal
- Diagonals bisect each other at right angles
- Has 2 lines of symmetry

Formulas:

- Perimeter = 4 × side
- Area = ½ × d_1 × d_2 (where d_1 and d_2 are diagonals)

5. TRAPEZIUM

Properties:

- One pair of opposite sides is parallel
- The parallel sides are called bases
- The non-parallel sides are called legs

Formulas:

- Area = ½ × (sum of parallel sides) × height
- Area = ½ × (a + b) × h

Example: A trapezium has parallel sides of 8 cm and 12 cm, and height 5 cm. Find its area.

- Area = ½ × (8 + 12) × 5
- Area = ½ × 20 × 5
- Area = **50 cm²**

6. KITE

Properties:

- Two pairs of adjacent sides are equal
- One pair of opposite angles are equal
- Diagonals intersect at right angles
- Has 1 line of symmetry

SUMMARY OF QUADRILATERAL PROPERTIES

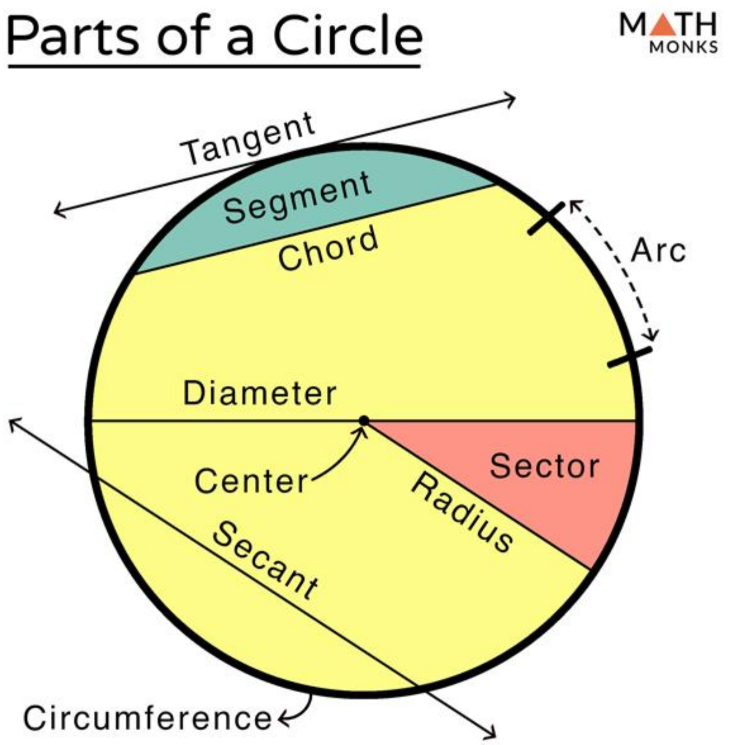
Quadrilateral	**Equal Sides**	**Right Angles**	**Parallel Sides**
Square	All 4	All 4	Both pairs
Rectangle	Opposite	All 4	Both pairs
Parallelogram	Opposite	None	Both pairs
Rhombus	All 4	None	Both pairs
Trapezium	None	None	One pair

Kite	Adjacent pairs	None	None

CIRCLES

A circle is a plane shape consisting of all points that are the same distance from a fixed point called the center.

PARTS OF A CIRCLE



1. CENTER: The fixed point in the middle of the circle (usually marked O)

2. RADIUS: The distance from the center to any point on the circle

- Symbol: r
- All radii of the same circle are equal

3. DIAMETER: A straight line passing through the center and touching two points on the circle

- Symbol: d
- Diameter = 2 × radius (d = 2r)

- The diameter is the longest chord

4. CHORD: A straight line joining any two points on the circle

5. ARC: A part of the circumference of a circle

6. CIRCUMFERENCE: The distance around the circle (perimeter of a circle)

- Formula: $C = 2\pi r$ or $C = \pi d$
- where π (pi) ≈ 3.14 or $22/7$

7. SECTOR: The region bounded by two radii and an arc

8. SEGMENT: The region bounded by a chord and an arc

9. TANGENT: A line that touches the circle at exactly one point

FORMULAS FOR CIRCLES

Circumference:

- $C = 2\pi r$ or $C = \pi d$
- where $\pi \approx 3.14$ or $22/7$

Area:

- $A = \pi r^2$

Example 1: A circle has radius 7 cm. Find its circumference. (Use $\pi = 22/7$)

Solution:

- $C = 2\pi r$
- $C = 2 \times 22/7 \times 7$
- $C = 2 \times 22$
- C = **44 cm**

Example 2: Find the area of a circle with radius 14 cm. (Use $\pi = 22/7$)

Solution:

- $A = \pi r^2$
- $A = 22/7 \times 14 \times 14$
- $A = 22/7 \times 196$
- $A = 22 \times 28$
- A = **616 cm²**

Example 3: A circle has diameter 20 cm. Find its circumference. (Use $\pi = 3.14$)

Solution:

- $C = \pi d$
- $C = 3.14 \times 20$
- C = **62.8 cm**

Example 4: If the circumference of a circle is 44 cm, find its radius. (Use $\pi = 22/7$)

Solution:

- $C = 2\pi r$
- $44 = 2 \times 22/7 \times r$
- $44 = 44r/7$
- $44 \times 7 = 44r$
- $308 = 44r$
- $r = 308/44$
- r = **7 cm**

RELATIONSHIP BETWEEN SHAPES

1. A square is a special type of rectangle (with all sides equal)

2. A square is a special type of rhombus (with right angles)
3. A rectangle is a special type of parallelogram (with right angles)
4. A rhombus is a special type of parallelogram (with all sides equal)

EVALUATION

1. What is the sum of angles in a triangle?
2. Find the perimeter of a rectangle with length 15 cm and width 8 cm
3. Calculate the area of a square with side 9 cm
4. A circle has radius 21 cm. Find its circumference (use $\pi = 22/7$)
5. Name two properties of a rhombus

ASSIGNMENT

1. Two angles of a triangle are 45° and 60°. Find the third angle
2. A rectangle has length 20 cm and width 12 cm. Find its area
3. Calculate the perimeter of a square with area 64 cm^2
4. Find the area of a circle with diameter 14 cm (use $\pi = 22/7$)
5. A trapezium has parallel sides 10 cm and 16 cm, and height 8 cm. Find its area

WEEK 3: THREE-DIMENSIONAL FIGURES

Learning Objectives

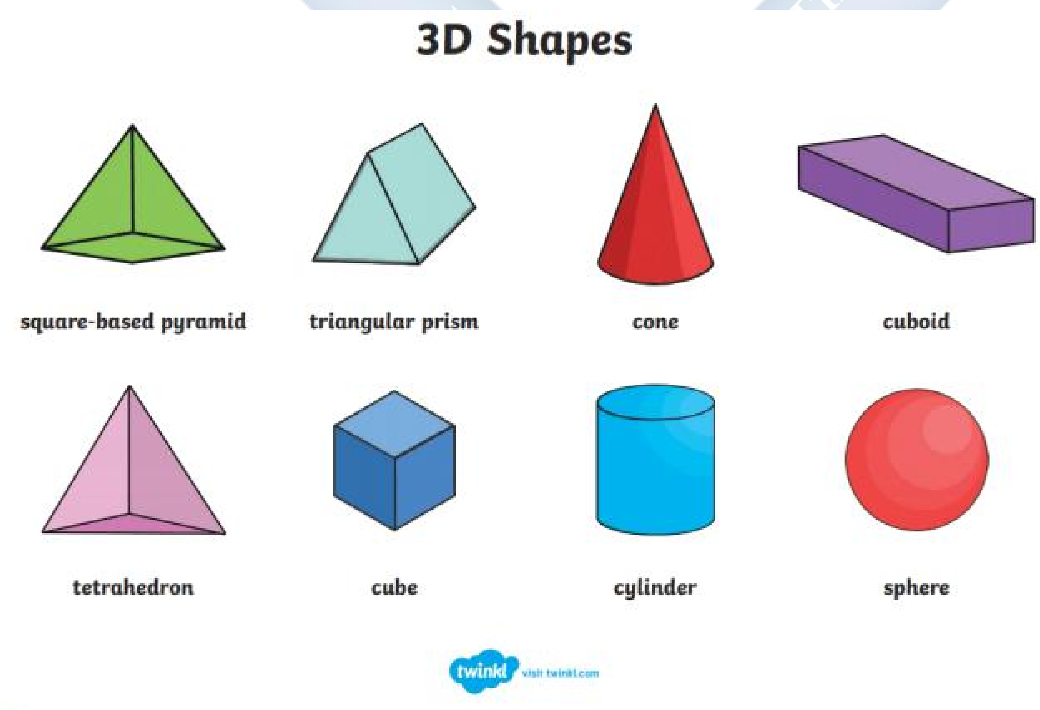
By the end of this lesson, students should be able to:

- Identify three-dimensional shapes
- State the properties of cubes and cuboids
- Describe cylinders, spheres, and cones
- Calculate surface areas and volumes
- Identify edges, faces, and vertices of 3D shapes

WHAT ARE THREE-DIMENSIONAL FIGURES?

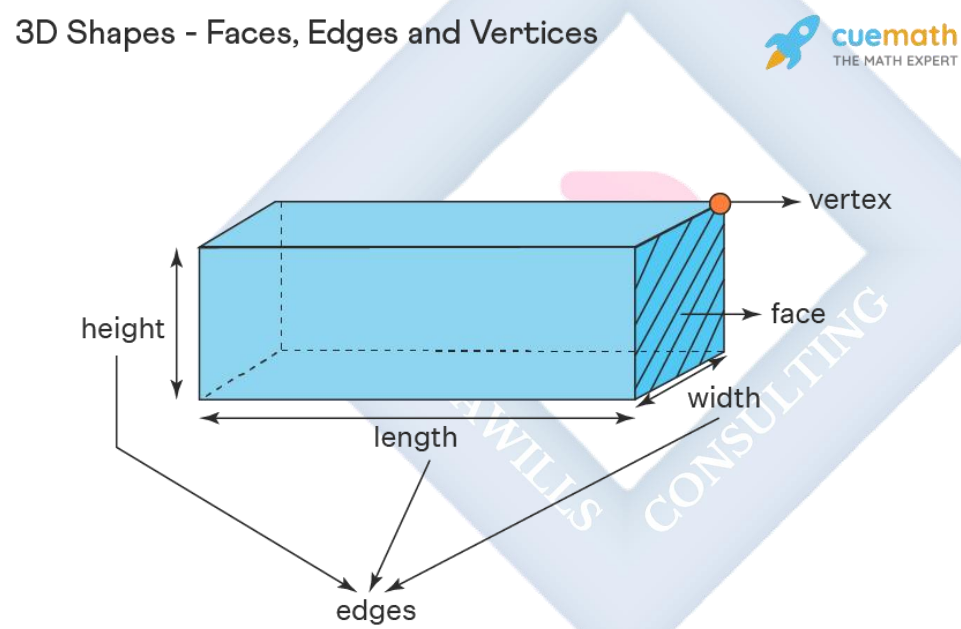
Three-dimensional figures (3D shapes or solids) are shapes that have three dimensions: length, width, and height (or depth). They have volume and take up space.

Common 3D shapes:



- Cube
- Cuboid (Rectangular prism)
- Cylinder
- Sphere
- Cone
- Pyramid

PARTS OF THREE-DIMENSIONAL SHAPES



1. FACE: A flat surface of a 3D shape

2. EDGE: A line segment where two faces meet

3. VERTEX (plural: VERTICES): A point where edges meet (corner)

CUBE

A cube is a three-dimensional shape with six identical square faces.

PROPERTIES OF A CUBE:

- 6 faces (all squares of equal size)
- 12 edges (all equal in length)
- 8 vertices
- All faces meet at right angles
- All edges are equal

FORMULAS:

Volume of a Cube:

- $V = \text{side} \times \text{side} \times \text{side} = s^3$
- where s = length of one side

Surface Area of a Cube:

- Surface Area = $6 \times (\text{side})^2 = 6s^2$
- (Because there are 6 identical square faces)

Example 1: A cube has sides of length 5 cm. Find its volume.

Solution:

- $V = s^3$
- $V = 5^3$
- $V = 5 \times 5 \times 5$
- V = **125 cm³**

Example 2: Find the surface area of a cube with side 4 cm.

Solution:

- Surface Area = $6s^2$

- $= 6 \times 4^2$
- $= 6 \times 16$
- **= 96 cm²**

Example 3: The volume of a cube is 216 cm^3. Find the length of its side.

Solution:

- $V = s^3$
- $216 = s^3$
- $s = \sqrt[3]{216}$
- s = **6 cm**

CUBOID (RECTANGULAR PRISM)

A cuboid is a three-dimensional shape with six rectangular faces.

PROPERTIES OF A CUBOID:

- 6 faces (all rectangles; opposite faces are equal)
- 12 edges (4 of each dimension)
- 8 vertices
- All faces meet at right angles

FORMULAS:

Volume of a Cuboid:

- V = length × width × height
- $V = l \times w \times h$

Surface Area of a Cuboid:

- Surface Area = $2(lw + lh + wh)$

- Or = 2(lb + bh + lh) where l=length, b=breadth, h=height

Example 1: A cuboid has length 8 cm, width 5 cm, and height 3 cm. Find its volume.

Solution:

- $V = l \times w \times h$
- $V = 8 \times 5 \times 3$
- V = **120 cm³**

Example 2: Find the surface area of a cuboid with length 10 cm, width 6 cm, and height 4 cm.

Solution:

- Surface Area = 2(lw + lh + wh)
- $= 2(10\times6 + 10\times4 + 6\times4)$
- $= 2(60 + 40 + 24)$
- $= 2(124)$
- = **248 cm²**

Example 3: A rectangular box has volume 360 cm³. If its length is 10 cm and width is 6 cm, find its height.

Solution:

- $V = l \times w \times h$
- $360 = 10 \times 6 \times h$
- $360 = 60h$
- $h = 360/60$
- h = **6 cm**

CYLINDER

A cylinder is a three-dimensional shape with two parallel circular bases connected by a curved surface.

PROPERTIES OF A CYLINDER:

- 2 circular faces (top and bottom bases)
- 1 curved surface
- 2 edges (circular edges at top and bottom)
- No vertices

Parts of a Cylinder:

- **Radius (r):** Radius of the circular base
- **Height (h):** Distance between the two circular bases

FORMULAS:

Volume of a Cylinder:

- $V = \pi r^2 h$
- where r = radius, h = height

Curved Surface Area:

- $CSA = 2\pi rh$

Total Surface Area:

- $TSA = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$
- (Two circular bases + curved surface)

Example 1: A cylinder has radius 7 cm and height 10 cm. Find its volume. (Use $\pi = 22/7$)

Solution:

- $V = \pi r^2 h$
- $V = 22/7 \times 7^2 \times 10$

- $V = 22/7 \times 49 \times 10$
- $V = 22 \times 7 \times 10$
- V = **1,540 cm^3**

Example 2: Find the curved surface area of a cylinder with radius 3.5 cm and height 12 cm. (Use $\pi = 22/7$)

Solution:

- $CSA = 2\pi rh$
- $= 2 \times 22/7 \times 3.5 \times 12$
- $= 2 \times 22/7 \times 42$
- $= 2 \times 22 \times 6$
- **= 264 cm^2**

Example 3: A cylindrical tank has radius 14 cm and height 20 cm. Find its total surface area. (Use $\pi = 22/7$)

Solution:

- $TSA = 2\pi r(r + h)$
- $= 2 \times 22/7 \times 14 \times (14 + 20)$
- $= 2 \times 22 \times 2 \times 34$
- $= 44 \times 34$
- **= 2,992 cm^2**

SPHERE

A sphere is a perfectly round three-dimensional shape. All points on the surface are the same distance from the center.

PROPERTIES OF A SPHERE:

- 1 curved surface (no flat faces)
- No edges
- No vertices
- Every point on the surface is equidistant from the center

Examples: Ball, globe, orange

Parts of a Sphere:

- **Radius (r):** Distance from center to any point on the surface
- **Diameter (d):** Distance across the sphere through the center ($d = 2r$)

FORMULAS:

Volume of a Sphere:

- $V = 4/3\ \pi r^3$

Surface Area of a Sphere:

- $SA = 4\pi r^2$

Example 1: Find the volume of a sphere with radius 3 cm. (Use $\pi = 3.14$)

Solution:

- $V = 4/3\ \pi r^3$
- $V = 4/3 \times 3.14 \times 3^3$
- $V = 4/3 \times 3.14 \times 27$
- $V = 4/3 \times 84.78$
- $V = 4 \times 28.26$
- $V = \mathbf{113.04\ cm^3}$

Example 2: A ball has radius 7 cm. Find its surface area. (Use $\pi = 22/7$)

Solution:

- $SA = 4\pi r^2$
- $= 4 \times 22/7 \times 7^2$
- $= 4 \times 22/7 \times 49$
- $= 4 \times 22 \times 7$
- $= \mathbf{616\ cm^2}$

CONE

A cone is a three-dimensional shape with a circular base and a curved surface that tapers to a point called the apex or vertex.

PROPERTIES OF A CONE:

- 1 circular flat face (base)
- 1 curved surface
- 1 edge (circular edge of the base)
- 1 vertex (apex/tip)

Parts of a Cone:

- **Radius (r):** Radius of the circular base
- **Height (h):** Perpendicular distance from base to apex
- **Slant height (l):** Distance from any point on the circular edge to the apex

Relationship: $l^2 = r^2 + h^2$ (Pythagoras)

FORMULAS:

Volume of a Cone:

- $V = 1/3\ \pi r^2 h$

Curved Surface Area:

- $CSA = \pi rl$
- where l = slant height

Total Surface Area:

- $TSA = \pi r^2 + \pi rl = \pi r(r + l)$

Example 1: A cone has radius 7 cm and height 24 cm. Find its volume. (Use $\pi = 22/7$)

Solution:

- $V = 1/3\ \pi r^2 h$
- $V = 1/3 \times 22/7 \times 7^2 \times 24$
- $V = 1/3 \times 22/7 \times 49 \times 24$
- $V = 1/3 \times 22 \times 7 \times 24$
- $V = 1/3 \times 3{,}696$
- $V =$ **1,232 cm³**

Example 2: Find the slant height of a cone with radius 5 cm and height 12 cm.

Solution:

- $l^2 = r^2 + h^2$
- $l^2 = 5^2 + 12^2$
- $l^2 = 25 + 144$
- $l^2 = 169$
- $l = \sqrt{169}$
- $l =$ **13 cm**

PYRAMID

A pyramid is a three-dimensional shape with a polygon base and triangular faces that meet at a point (apex).

PROPERTIES OF A PYRAMID:

- 1 polygon base (can be triangle, square, pentagon, etc.)
- Triangular faces
- Edges connecting base vertices to apex
- 1 apex (vertex at the top)

Types:

- **Triangular pyramid** (base is a triangle) - also called tetrahedron
- **Square pyramid** (base is a square)
- **Pentagonal pyramid** (base is a pentagon)

For a Square Pyramid:

Volume:

- $V = 1/3 \times \text{base area} \times \text{height}$
- $V = 1/3 \times s^2 \times h$ (where s = side of square base)

Example: A square pyramid has base side 6 cm and height 8 cm. Find its volume.

Solution:

- $V = 1/3 \times s^2 \times h$
- $V = 1/3 \times 6^2 \times 8$
- $V = 1/3 \times 36 \times 8$
- $V = 1/3 \times 288$
- V = **96 cm³**

SUMMARY TABLE OF 3D SHAPES

Shape	Faces	Edges	Vertices
Cube	6	12	8
Cuboid	6	12	8
Cylinder	2 (circular) + 1 curved	2	0
Sphere	1 (curved)	0	0
Cone	1 (circular) + 1 curved	1	1
Square Pyramid	5 (1 square + 4 triangles)	8	5

REAL-LIFE EXAMPLES

Cube: Dice, Rubik's cube, sugar cubes

Cuboid: Boxes, books, bricks, rooms

Cylinder: Cans, pipes, drums, pillars

Sphere: Ball, globe, marbles, oranges

Cone: Ice cream cone, traffic cone, party hat

Pyramid: Egyptian pyramids, tent

EVALUATION

1. How many faces does a cube have?
2. Find the volume of a cuboid with length 12 cm, width 8 cm, and height 5 cm
3. A cylinder has radius 7 cm and height 15 cm. Find its volume (use $\pi = 22/7$)
4. Calculate the surface area of a cube with side 10 cm
5. What is the formula for the volume of a sphere?

ASSIGNMENT

1. A cube has volume 343 cm^3. Find the length of its side
2. Find the surface area of a cuboid with dimensions 10 cm × 6 cm × 4 cm
3. A sphere has radius 21 cm. Find its surface area (use $\pi = 22/7$)
4. Calculate the volume of a cone with radius 6 cm and height 8 cm (use $\pi = 3.14$)
5. How many vertices does a cylinder have?

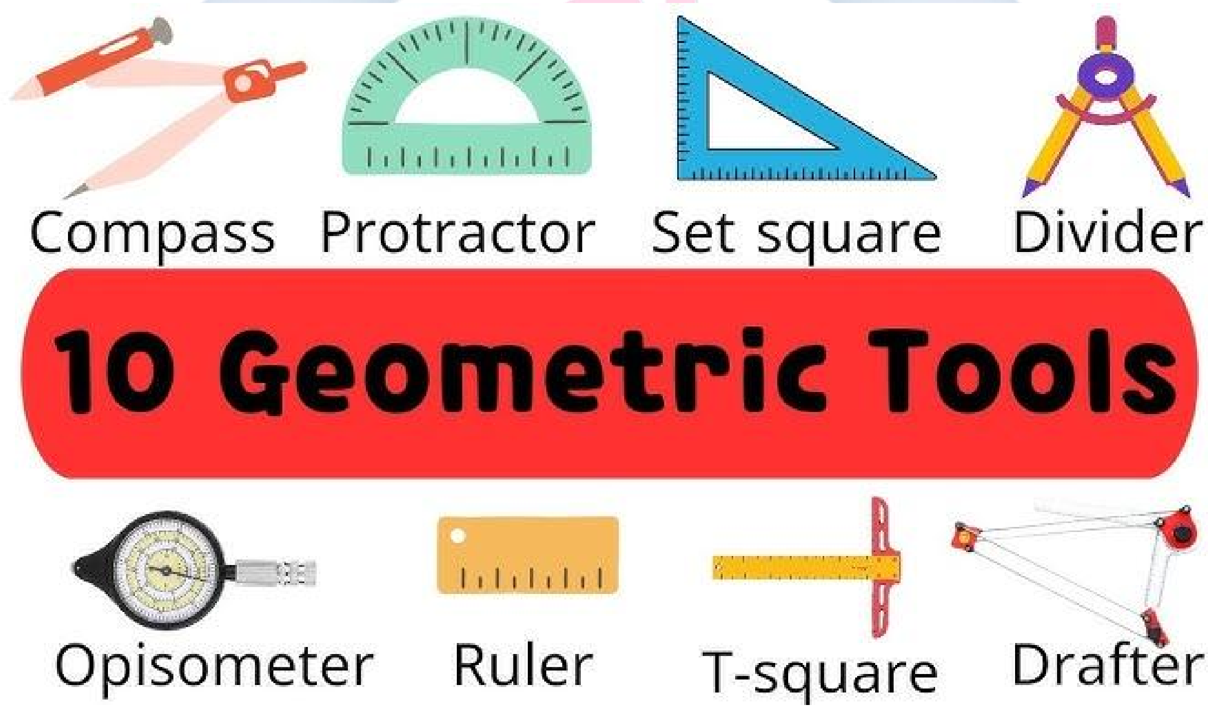
WEEK 4: CONSTRUCTIONS

Learning Objectives

By the end of this lesson, students should be able to:

- Use mathematical instruments properly (ruler, compass, protractor)
- Construct line segments of given lengths
- Bisect line segments
- Construct angles of specific measures
- Bisect angles
- Construct perpendicular lines

MATHEMATICAL INSTRUMENTS FOR CONSTRUCTION



1. RULER (Straight Edge):

- Used to draw straight lines

- Used to measure lengths
- Has markings in centimeters and millimeters

2. COMPASS (Pair of Compasses):

- Used to draw circles and arcs
- Used to transfer measurements
- Has two legs: one with a needle point, one with a pencil

3. PROTRACTOR:

- Used to measure angles
- Used to draw angles of specific sizes
- Marked from 0° to 180° (usually two scales)

4. SET SQUARES:

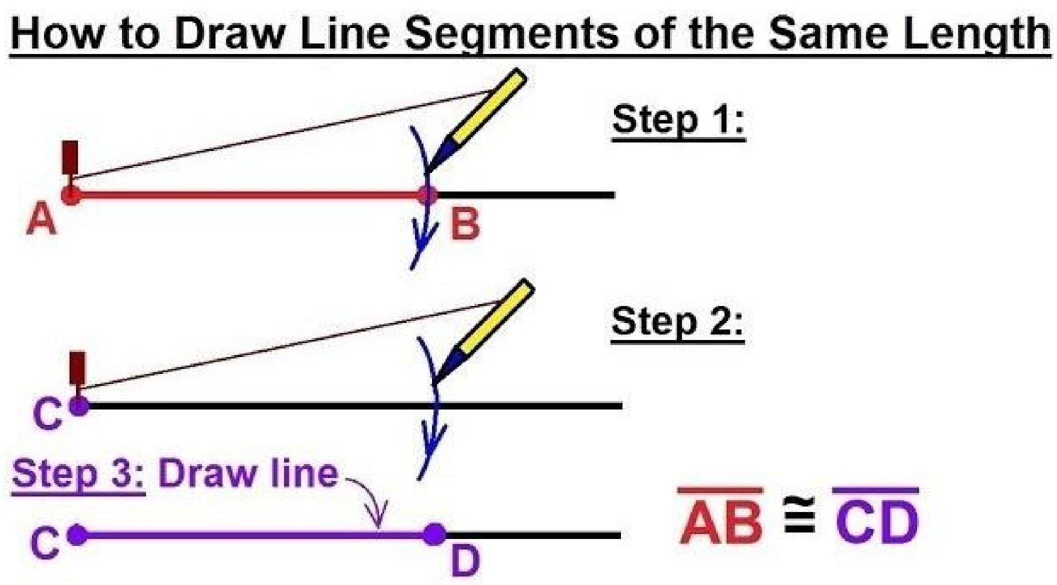
- Triangular instruments
- Used to draw perpendicular and parallel lines
- Common angles: 30°-60°-90° and 45°-45°-90°

5. PENCIL:

- Sharp pencil for accurate constructions
- Use HB or H pencil for construction lines

BASIC CONSTRUCTION TECHNIQUES

CONSTRUCTION 1: DRAWING A LINE SEGMENT OF GIVEN LENGTH

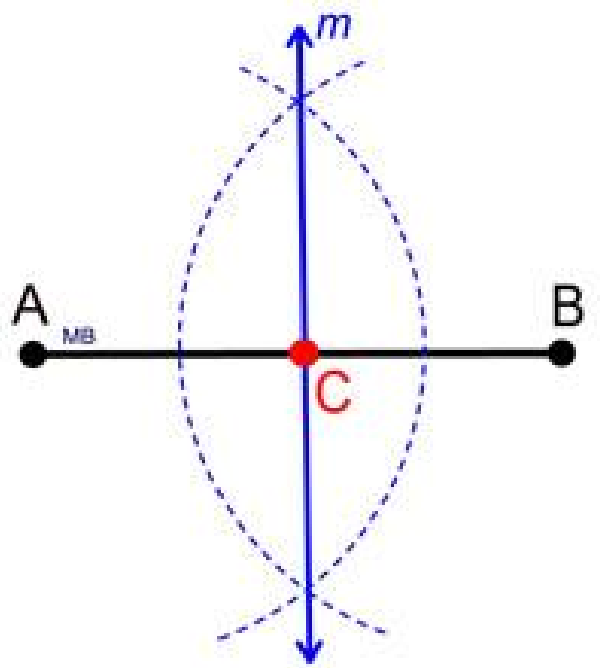


To draw a line segment AB of length 6 cm:

Steps:

1. Mark a point A on your paper
2. Place the ruler with 0 cm at point A
3. Mark point B at 6 cm
4. Join A and B with a straight line
5. Label the line segment AB = 6 cm

CONSTRUCTION 2: BISECTING A LINE SEGMENT



To bisect means to divide into two equal parts.

To bisect line segment AB:

Steps:

1. Draw line segment AB
2. With center A and radius more than half of AB, draw arcs above and below AB
3. With the same radius and center B, draw arcs to intersect the previous arcs at points P and Q
4. Join P and Q with a straight line
5. The line PQ bisects AB at point M
6. AM = MB (M is the midpoint)

Note: PQ is also perpendicular to AB

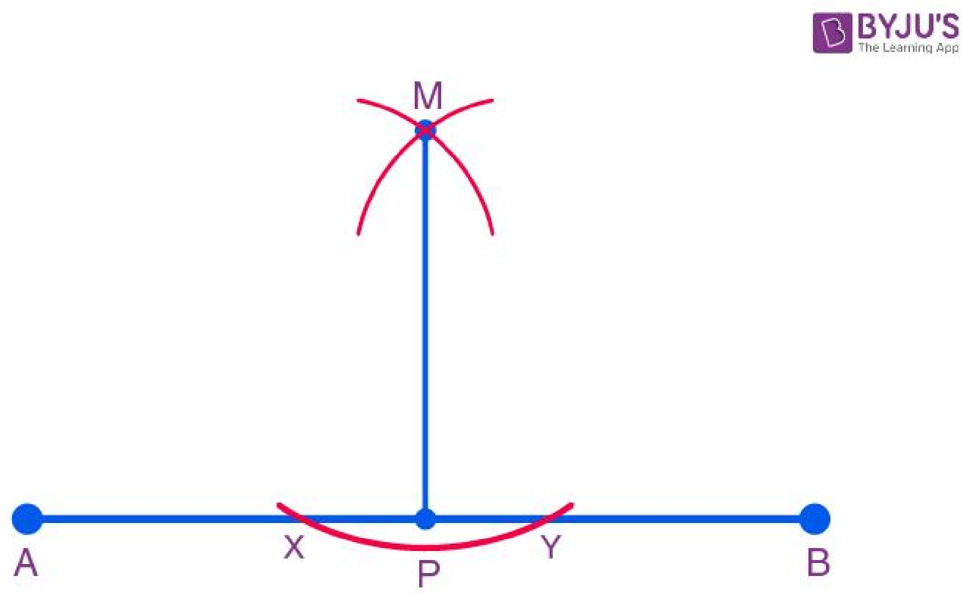
Example: Bisect a line segment of length 8 cm.

Steps:

1. Draw AB = 8 cm

2. Open compass to more than 4 cm (more than half of 8 cm)
3. With center A, draw arcs above and below AB
4. With same radius and center B, draw arcs cutting previous arcs
5. Join the intersection points
6. The midpoint M divides AB into two equal parts of 4 cm each

CONSTRUCTION 3: CONSTRUCTING PERPENDICULAR LINES



METHOD 1: Using a Set Square

To draw a line perpendicular to AB at point P:

Steps:

1. Draw line AB and mark point P on it
2. Place the set square with one edge along AB
3. Slide it until the perpendicular edge passes through P
4. Draw a line along the perpendicular edge

5. This line is perpendicular to AB at P

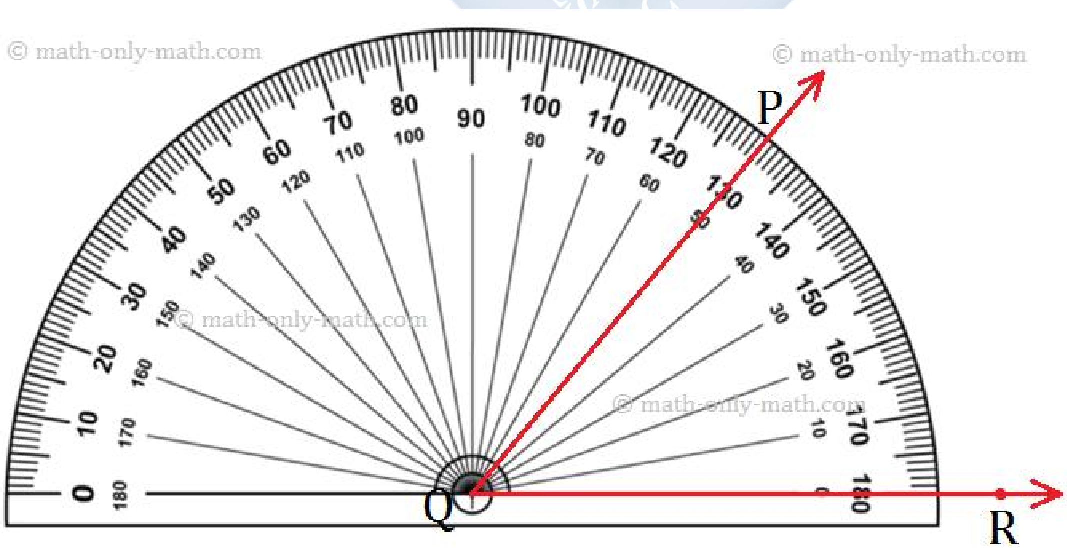
METHOD 2: Using Compass (Perpendicular at the End of a Line)

To construct a perpendicular to AB at point B:

Steps:

1. Draw line AB
2. With center B, draw an arc cutting AB at point C
3. With center C and same radius, draw an arc
4. With center C and larger radius, draw an arc above AB
5. From the previous arc intersection and with same radius, draw another arc to intersect the last arc at D
6. Join BD
7. BD is perpendicular to AB

CONSTRUCTION 4: CONSTRUCTING ANGLES USING A PROTRACTOR



To construct an angle of 60°:

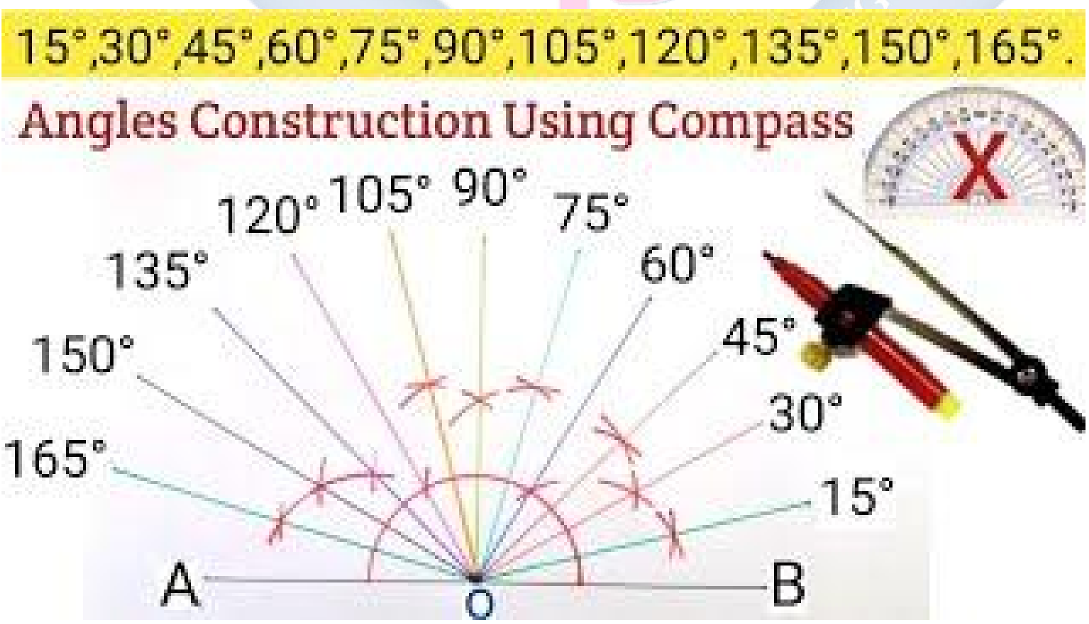
Steps:

1. Draw a horizontal line AB
2. Place the protractor with center at A and baseline along AB
3. Mark a point C at 60° on the protractor
4. Remove the protractor
5. Join A and C
6. ∠BAC = 60°

Common Angles to Practice:

- 30°, 45°, 60°, 90°, 120°, 135°, 150°

CONSTRUCTION 5: CONSTRUCTING ANGLES USING COMPASS



To construct an angle of 60° using compass:

Steps:

1. Draw a line AB
2. With center A and any convenient radius, draw an arc cutting AB at point P
3. With same radius and center P, draw an arc cutting the first arc at Q
4. Join A and Q
5. ∠BAQ = 60°

To construct an angle of 90° using compass:

Steps:

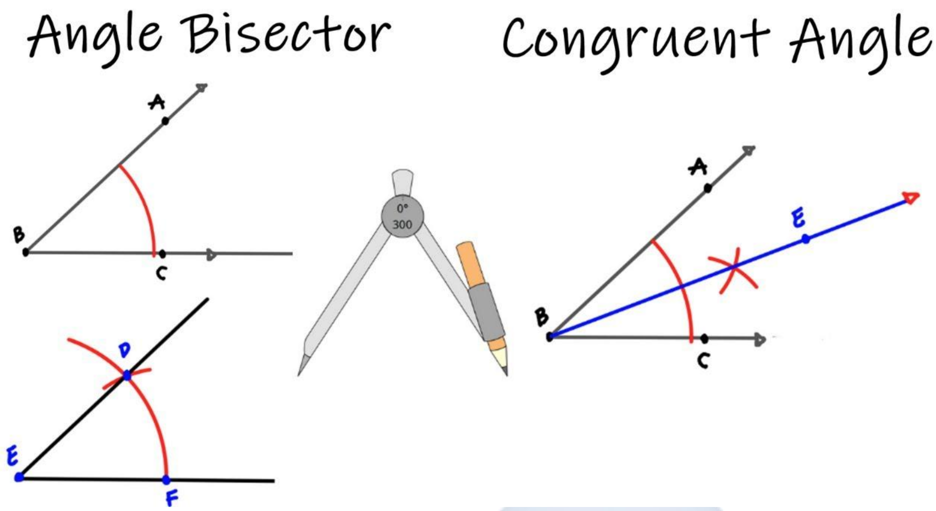
1. Draw a line AB
2. With center A, draw an arc cutting AB at P
3. With center P and same radius, draw arc cutting first arc at Q
4. With center Q and same radius, draw arc cutting first arc at R
5. With centers Q and R and larger equal radii, draw two arcs intersecting at S
6. Join AS
7. ∠BAS = 90°

To construct an angle of 120° using compass:

Steps:

1. Follow steps for 60° to get point Q
2. With same radius and center Q, draw another arc cutting the first arc at R
3. Join AR
4. ∠BAR = 120°

CONSTRUCTION 6: BISECTING AN ANGLE



To bisect ∠ABC:

Steps:

1. Draw angle ABC
2. With center B and any radius, draw an arc cutting BA at P and BC at Q
3. With centers P and Q and equal radii (more than half of PQ), draw two arcs intersecting at point R
4. Join BR
5. BR bisects ∠ABC
6. ∠ABR = ∠RBC

Example: Bisect an angle of 80°.

Result: You get two angles of 40° each.

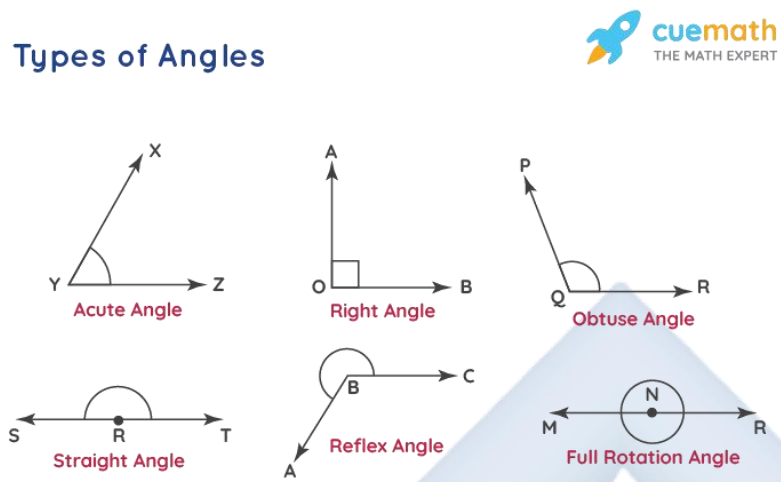
Example: Construct an angle of 45° using angle bisection.

Steps:

1. First construct a 90° angle
2. Bisect the 90° angle

3. The result is 45°

CONSTRUCTION 7: CONSTRUCTING SPECIAL ANGLES



To construct 30°:

1. Construct 60° angle
2. Bisect the 60° angle
3. Result: 30°

To construct 45°:

1. Construct 90° angle
2. Bisect the 90° angle
3. Result: 45°

To construct 15°:

1. Construct 30° angle
2. Bisect the 30° angle
3. Result: 15°

To construct 75°:

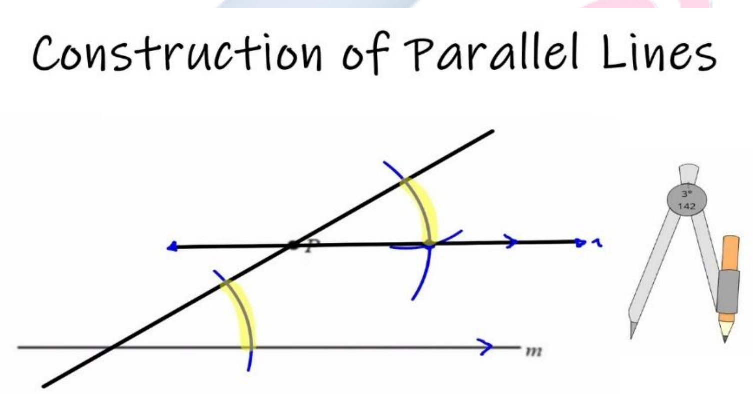
1. Construct 90° angle

2. Construct 60° angle from the same baseline
3. The angle between them is 30°
4. Bisect the 30° angle to get 15°
5. 60° + 15° = 75°

OR

1. Construct 60° angle
2. Bisect to get 30°
3. Bisect 30° to get 15°
4. 60° + 15° = 75°

CONSTRUCTION 8: DRAWING PARALLEL LINES



METHOD 1: Using Set Squares

Steps:

1. Draw line AB
2. Place one set square with an edge along AB
3. Place second set square against another edge of first
4. Slide first set square along second

5. Draw a line along the edge
6. This line is parallel to AB

METHOD 2: Using Compass

To draw a line parallel to AB through point P:

Steps:

1. Draw any line through P cutting AB at Q
2. At P, construct an angle equal to the angle at Q (alternate angles)
3. This creates a parallel line

CONSTRUCTION TIPS

1. **Keep your pencil sharp** for accurate constructions
2. **Don't press too hard** - construction lines should be light
3. **Don't erase construction arcs** - they show your method
4. **Label all points clearly**
5. **Keep your compass tight** so the radius doesn't change
6. **Practice regularly** to improve accuracy

PRACTICAL APPLICATIONS

1. Architecture: Designing buildings with precise angles and measurements

2. Engineering: Creating technical drawings and blueprints

3. Art: Creating symmetrical designs and patterns

4. Carpentry: Making accurate cuts and joints

5. Navigation: Drawing maps and determining directions

EVALUATION

1. Name three instruments used for geometric constructions
2. Describe how to bisect a line segment
3. How can you construct a 90° angle using a compass?
4. What is the first step in bisecting an angle?
5. How do you construct a 45° angle?

ASSIGNMENT

1. Draw a line segment of 7 cm and bisect it
2. Construct an angle of 60° using a compass
3. Bisect an angle of 100°. What is the measure of each resulting angle?
4. Describe the steps to construct a perpendicular line
5. How can you construct a 30° angle? (Describe two methods)

WEEK 6: ANGLES

Learning Objectives

By the end of this lesson, students should be able to:

- Define an angle
- Identify different types of angles
- Measure angles using a protractor
- Calculate unknown angles
- Apply angle properties of triangles and quadrilaterals

WHAT IS AN ANGLE?

An angle is formed when two straight lines (called arms or rays) meet at a point (called the vertex).

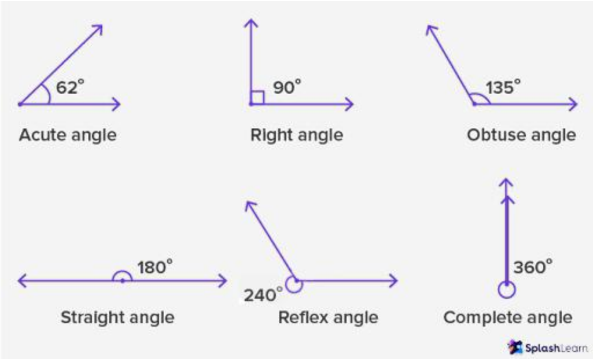
Parts of an Angle:

- **Vertex:** The point where the two lines meet
- **Arms:** The two straight lines forming the angle
- **Notation:** ∠ABC or ∠B (where B is the vertex)

Measuring Angles:

- Angles are measured in degrees (°)
- Symbol: °
- We use a protractor to measure angles

TYPES OF ANGLES BY SIZE



1. ACUTE ANGLE

- An angle less than 90°
- Range: 0° < angle < 90°
- Examples: 30°, 45°, 60°, 75°

2. RIGHT ANGLE

- An angle equal to exactly 90°
- Shown by a small square at the vertex
- Found in squares, rectangles, and perpendicular lines

3. OBTUSE ANGLE

- An angle greater than 90° but less than 180°
- Range: 90° < angle < 180°
- Examples: 100°, 120°, 135°, 160°

4. STRAIGHT ANGLE

- An angle equal to exactly 180°
- Forms a straight line
- The two arms point in opposite directions

5. REFLEX ANGLE

- An angle greater than 180° but less than 360°
- Range: 180° < angle < 360°
- Examples: 200°, 270°, 315°

6. COMPLETE ANGLE (REVOLUTION)

- An angle equal to exactly 360°
- A full turn or rotation
- Returns to starting position

Summary Table:

Type	Measure
Acute	0° to 90° (exclusive)
Right	Exactly 90°
Obtuse	90° to 180° (exclusive)
Straight	Exactly 180°
Reflex	180° to 360° (exclusive)
Complete	Exactly 360°

ANGLE RELATIONSHIPS

1. COMPLEMENTARY ANGLES

- Two angles that add up to 90°
- They are complements of each other

Examples:

- 30° and 60° are complementary (30° + 60° = 90°)
- 45° and 45° are complementary (45° + 45° = 90°)
- 25° and 65° are complementary (25° + 65° = 90°)

Example 1: Find the complement of 35°.

Solution:

- Complement = 90° - 35° = **55°**

Example 2: Two complementary angles are in the ratio 2:3. Find the angles.

Solution:

- Let the angles be 2x and 3x
- 2x + 3x = 90°
- 5x = 90°
- x = 18°
- Angles are: 2(18°) = 36° and 3(18°) = 54°

2. SUPPLEMENTARY ANGLES

- Two angles that add up to 180°
- They are supplements of each other

Examples:

- 110° and 70° are supplementary (110° + 70° = 180°)
- 120° and 60° are supplementary
- 90° and 90° are supplementary

Example 1: Find the supplement of 125°.

Solution:

- Supplement = 180° - 125° = **55°**

Example 2: Two supplementary angles differ by 40°. Find the angles.

Solution:

- Let the angles be x and (x + 40°)

- $x + (x + 40°) = 180°$
- $2x + 40° = 180°$
- $2x = 140°$
- $x = 70°$
- Angles are: 70° and 110°

ANGLES FORMED BY INTERSECTING LINES

VERTICALLY OPPOSITE ANGLES

When two straight lines intersect, they form four angles. The angles opposite each other are called vertically opposite angles, and they are equal.

Properties:

- $\angle 1 = \angle 3$ (vertically opposite)
- $\angle 2 = \angle 4$ (vertically opposite)
- $\angle 1 + \angle 2 = 180°$ (supplementary - on a straight line)
- $\angle 1 + \angle 4 = 180°$ (supplementary - on a straight line)

Example: Two lines intersect. If one angle is 65°, find the other three angles.

Solution:

- Vertically opposite angle = 65°
- Adjacent angles = 180° - 65° = 115° (each)
- The four angles are: **65°, 115°, 65°, 115°**

ANGLES ON A STRAIGHT LINE

Property: Angles on a straight line add up to 180°

Example 1: Two angles on a straight line are 110° and x. Find x.

Solution:

- 110° + x = 180°
- x = 180° - 110°
- x = **70°**

Example 2: Three angles on a straight line are 50°, 75°, and x. Find x.

Solution:

- 50° + 75° + x = 180°
- 125° + x = 180°
- x = **55°**

ANGLES AT A POINT

Property: Angles around a point add up to 360°

Example 1: Four angles meet at a point: 80°, 90°, 120°, and x. Find x.

Solution:

- 80° + 90° + 120° + x = 360°
- 290° + x = 360°
- x = **70°**

ANGLES IN PARALLEL LINES

When a straight line (called a transversal) crosses two parallel lines, several angle relationships are formed:

1. CORRESPONDING ANGLES

- Angles in the same relative position at each intersection

- Corresponding angles are **equal**
- Also called F-angles (they form an F shape)

2. ALTERNATE ANGLES

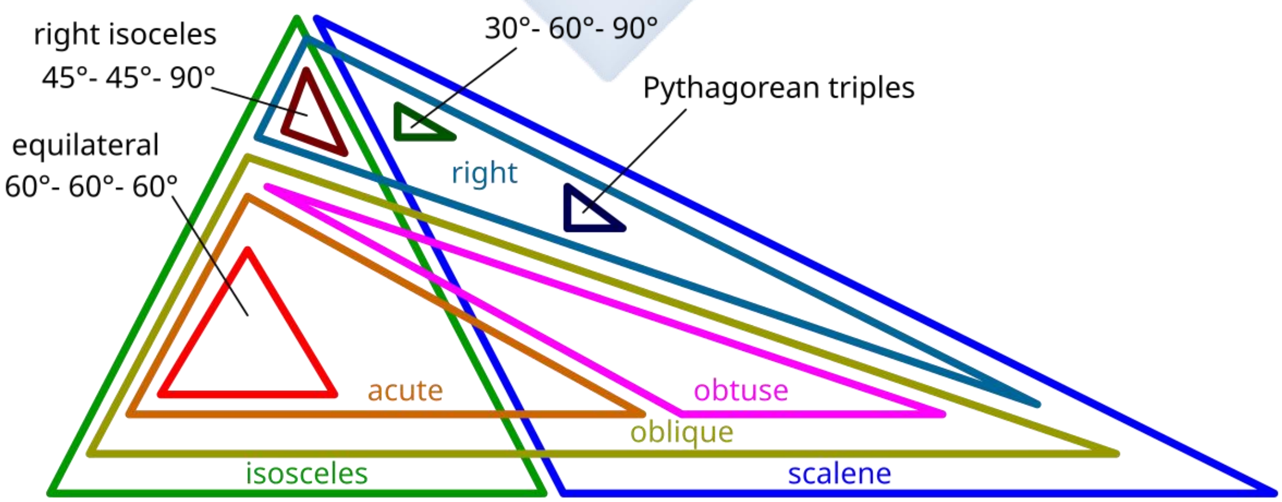
- Angles on opposite sides of the transversal, between the parallel lines
- Alternate angles are **equal**
- Also called Z-angles (they form a Z shape)

3. INTERIOR ANGLES (Co-interior angles)

- Angles on the same side of the transversal, between the parallel lines
- Interior angles are **supplementary** (add up to 180°)
- Also called C-angles (they form a C shape)

Example: Two parallel lines are cut by a transversal. If one angle is 65°, find: a) The corresponding angle b) The alternate angle c) The co-interior angle

Solution: a) Corresponding angle = **65°** (equal) b) Alternate angle = **65°** (equal) c) Co-interior angle = 180° - 65° = **115°** (supplementary)



ANGLE PROPERTIES OF TRIANGLES

1. ANGLE SUM PROPERTY

The sum of the three interior angles of any triangle is 180°.

$\angle A + \angle B + \angle C = 180°$

Example 1: In triangle ABC, $\angle A = 50°$ and $\angle B = 70°$. Find $\angle C$.

Solution:

- $\angle A + \angle B + \angle C = 180°$
- $50° + 70° + \angle C = 180°$
- $120° + \angle C = 180°$
- $\angle C =$ **60°**

Example 2: The angles of a triangle are in the ratio 2:3:4. Find each angle.

Solution:

- Let the angles be $2x$, $3x$, and $4x$
- $2x + 3x + 4x = 180°$
- $9x = 180°$
- $x = 20°$
- Angles are: $2(20°) = 40°$, $3(20°) = 60°$, $4(20°) = 80°$

2. EXTERIOR ANGLE PROPERTY

An exterior angle of a triangle equals the sum of the two opposite interior angles.

Example: In a triangle, two interior angles are 40° and 75°. Find the exterior angle at the third vertex.

Solution:

- Exterior angle = 40° + 75° = **115°**

OR

- Third interior angle = 180° - (40° + 75°) = 65°
- Exterior angle = 180° - 65° = **115°**

3. SPECIAL TRIANGLES

Equilateral Triangle:

- All three angles are equal
- Each angle = 180° ÷ 3 = 60°

Isosceles Triangle:

- Two angles are equal (base angles)
- If base angles are each x°, then:
- 2x + third angle = 180°

Example: In an isosceles triangle, each base angle is 55°. Find the third angle.

Solution:

- Let third angle = y
- 55° + 55° + y = 180°
- 110° + y = 180°
- y = **70°**

Right-Angled Triangle:

- One angle is 90°
- The other two angles add up to 90° (complementary)

Example: One acute angle in a right-angled triangle is 35°. Find the other acute angle.

Solution:

- 90° + 35° + other angle = 180°
- 125° + other angle = 180°
- Other angle = **55°**

OR simply: 90° - 35° = **55°**

ANGLE PROPERTIES OF QUADRILATERALS

1. ANGLE SUM PROPERTY

The sum of the four interior angles of any quadrilateral is 360°.

∠A + ∠B + ∠C + ∠D = 360°

Example 1: Three angles of a quadrilateral are 80°, 110°, and 95°. Find the fourth angle.

Solution:

- Sum = 360°
- 80° + 110° + 95° + fourth angle = 360°
- 285° + fourth angle = 360°
- Fourth angle = **75°**

Example 2: In a quadrilateral, the angles are in the ratio 2:3:4:6. Find each angle.

Solution:

- Let angles be 2x, 3x, 4x, and 6x
- 2x + 3x + 4x + 6x = 360°
- 15x = 360°
- x = 24°

- Angles are:
    - $2(24°) = 48°$
    - $3(24°) = 72°$
    - $4(24°) = 96°$
    - $6(24°) = 144°$

2. SPECIAL QUADRILATERALS

Square:

- All four angles = 90°
- $4 \times 90° = 360°$ ✓

Rectangle:

- All four angles = 90°
- $4 \times 90° = 360°$ ✓

Parallelogram:

- Opposite angles are equal
- Adjacent angles are supplementary (add up to 180°)
- If one angle is x°, opposite angle is also x°
- Adjacent angles are (180° - x°) each

Example: In a parallelogram, one angle is 65°. Find the other three angles.

Solution:

- Opposite angle = 65°
- Adjacent angles = 180° - 65° = 115° (each)
- The four angles are: **65°, 115°, 65°, 115°**

Rhombus:

- Same angle properties as parallelogram
- Opposite angles are equal
- Adjacent angles are supplementary

Trapezium:

- Sum of all angles = 360°
- If one pair of sides is parallel:
  - Co-interior angles on each side are supplementary

Kite:

- One pair of opposite angles are equal
- Sum of all angles = 360°

SOLVING ANGLE PROBLEMS

Example 1: In the figure, AB is parallel to CD. If ∠1 = 115°, find ∠2.

Solution (if ∠2 is corresponding):

- ∠2 = 115° (corresponding angles are equal)

Solution (if ∠2 is alternate):

- ∠2 = 115° (alternate angles are equal)

Solution (if ∠2 is co-interior):

- ∠2 = 180° - 115° = 65°

Example 2: Calculate all the angles in the figure where two straight lines intersect, and one angle is 118°.

Solution:

- Vertically opposite to 118° = 118°
- Adjacent angles = 180° - 118° = 62° (each)
- Angles are: **118°, 62°, 118°, 62°**

EVALUATION

1. What is the complement of 42°?
2. Find the supplement of 135°
3. Two angles of a triangle are 55° and 75°. Find the third angle
4. The angles of a quadrilateral are 90°, 85°, 100°, and x. Find x
5. In a parallelogram, one angle is 70°. Find the adjacent angle

ASSIGNMENT

1. Classify these angles: (a) 45° (b) 135° (c) 90° (d) 200°
2. Two complementary angles are in the ratio 1:2. Find the angles
3. In triangle PQR, ∠P = 35° and ∠Q = 80°. Find ∠R
4. Three angles of a quadrilateral are 75°, 110°, and 85°. Find the fourth angle
5. In an isosceles triangle, the vertex angle is 40°. Find each base angle

WEEK 8: EVERYDAY STATISTICS I

Learning Objectives

By the end of this lesson, students should be able to:

- Define statistics
- Explain the need for statistics
- Identify where statistics is used in daily life
- Understand the importance of data
- Give examples of statistical information around us

WHAT IS STATISTICS?

Statistics is the branch of mathematics that deals with collecting, organizing, analyzing, presenting, and interpreting data (information).

Key Terms:

DATA: Facts, figures, or information collected for reference or analysis

- Examples: test scores, heights of students, daily temperatures, prices of goods

STATISTICS: The science of dealing with data

- Helps us make sense of large amounts of information
- Helps us see patterns and trends
- Helps us make decisions based on facts

THE NEED FOR STATISTICS

Statistics is needed for many important reasons:

1. TO ORGANIZE INFORMATION

Without statistics, we would have piles of unorganized numbers that mean nothing. Statistics helps us arrange data in a meaningful way.

Example: Instead of having a long list of test scores (78, 56, 89, 67, 92, 45, 88, 71, 80, 66...), statistics helps us organize this into:

- Highest score: 92
- Lowest score: 45
- Average score: 73
- Number of students who passed: 8 out of 10

2. TO MAKE COMPARISONS

Statistics allows us to compare different groups or situations.

Examples:

- Which class performed better in the exam?
- Is this year's harvest better than last year's?
- Which product sells more?
- Which treatment is more effective?

3. TO IDENTIFY PATTERNS AND TRENDS

Statistics helps us see patterns in data that we might miss otherwise.

Examples:

- Sales increasing every December (holiday season)
- More accidents happening on weekends
- Temperature rising over the years (climate change)
- Population growth over time

4. TO MAKE PREDICTIONS

Based on past data, statistics helps us predict future events.

Examples:

- Predicting next year's population
- Forecasting weather
- Estimating election results
- Planning inventory for a business

5. TO MAKE INFORMED DECISIONS

Statistics provides facts that help people, businesses, and governments make better decisions.

Examples:

- Government deciding where to build new schools based on population data
- A farmer deciding which crop to plant based on past yields
- A store deciding which products to stock based on sales data
- A doctor choosing the best treatment based on success rates

6. TO SOLVE PROBLEMS

Statistics helps identify and solve problems by showing what the real issues are.

Examples:

- Identifying subjects where students are struggling
- Finding out which roads have the most accidents
- Discovering which products are defective
- Locating areas with high disease rates

IMPORTANCE OF STATISTICS IN DAILY LIFE

Statistics affects almost every part of our daily lives, even when we don't realize it.

WHERE WE USE STATISTICS

1. IN EDUCATION

Uses:

- Recording student attendance
- Analyzing exam results
- Comparing performance across classes or schools
- Identifying subjects that need more attention
- Planning school budgets

Examples:

- "The average score in Mathematics is 65%"
- "80% of students passed the exam"
- "Attendance rate this term is 92%"

2. IN HEALTHCARE

Uses:

- Tracking disease outbreaks
- Recording patient data (temperature, blood pressure, etc.)
- Analyzing success rates of treatments
- Planning health programs
- Monitoring population health

Examples:

- "The normal body temperature is 37°C"
- "COVID-19 cases increased by 15% this week"
- "70% of patients recovered with this treatment"

- "Life expectancy in Nigeria is 54 years"

3. IN SPORTS

Uses:

- Recording player statistics (goals, runs, points)
- Analyzing team performance
- Comparing athletes
- Making predictions about game outcomes
- Selecting players

Examples:

- "Ronaldo has scored 800+ career goals"
- "The team won 75% of their matches"
- "Average attendance at matches is 45,000"

4. IN GOVERNMENT AND POLITICS

Uses:

- Conducting census (population count)
- Analyzing economic data
- Planning development projects
- Making policies
- Budgeting

Examples:

- "Nigeria's population is over 200 million"
- "Unemployment rate is 33%"
- "Inflation rate increased to 22%"

- "Government allocated ₦4 trillion to education"

5. IN BUSINESS AND ECONOMICS

Uses:

- Analyzing sales data
- Forecasting demand
- Setting prices
- Managing inventory
- Understanding customer behavior
- Market research

Examples:

- "Sales increased by 20% last quarter"
- "Most customers prefer product A"
- "Price of rice increased by ₦500 per bag"
- "The company made ₦10 million profit"

6. IN WEATHER AND CLIMATE

Uses:

- Recording daily temperatures
- Predicting weather patterns
- Tracking rainfall
- Studying climate change
- Planning agricultural activities

Examples:

- "Average temperature today is 28°C"

- "60% chance of rain tomorrow"
- "Annual rainfall is 1,500mm"
- "This is the hottest July in 50 years"

7. IN MEDIA AND NEWS

Uses:

- Reporting survey results
- Sharing election polls
- Presenting economic data
- Reporting crime statistics
- Covering sports statistics

Examples:

- "65% of Nigerians support the policy"
- "Crime rate decreased by 10%"
- "Viewership increased to 5 million"

8. IN EVERYDAY SHOPPING

Uses:

- Comparing prices
- Reading nutritional information
- Understanding discounts
- Checking product ratings
- Managing personal budgets

Examples:

- "This costs ₦500 per kg"

- "Save 20% on all items"
- "Contains 200 calories"
- "4.5 stars out of 5"

9. IN TRANSPORTATION

Uses:

- Recording traffic data
- Analyzing accident statistics
- Planning routes
- Managing public transport
- Monitoring fuel consumption

Examples:

- "Average speed on this road is 60 km/h"
- "50 accidents recorded this month"
- "Bus arrives every 15 minutes"

10. IN SCIENCE AND RESEARCH

Uses:

- Conducting experiments
- Analyzing results
- Testing hypotheses
- Drawing conclusions
- Sharing findings

Examples:

- "9 out of 10 scientists agree"

- "Success rate is 85%"
- "The experiment was repeated 100 times"

TYPES OF STATISTICAL DATA WE ENCOUNTER

1. NUMERICAL DATA (Numbers)

- Test scores: 75, 82, 90
- Prices: ₦500, ₦1,200, ₦850
- Heights: 1.5m, 1.7m, 1.6m
- Ages: 12, 15, 13, 14

2. CATEGORICAL DATA (Categories)

- Favorite colors: red, blue, green
- Gender: male, female
- Grades: A, B, C, D, F
- Days of the week: Monday, Tuesday, etc.

3. TIME-BASED DATA

- Daily temperature readings
- Monthly sales figures
- Yearly population growth
- Weekly attendance

EVALUATION

1. What is statistics?
2. Give three reasons why we need statistics

3. Name three areas where statistics is used in daily life
4. Why is it important for students to learn statistics?
5. How does statistics help in decision making?

ASSIGNMENT

1. Define data and give three examples
2. Explain how statistics is used in healthcare
3. List five ways you encounter statistics in your daily life
4. Why must statistical data be accurate?
5. Describe two ways statistics helps in education

WEEK 9: EVERYDAY STATISTICS II

Learning Objectives

By the end of this lesson, students should be able to:

- Understand different methods of collecting data
- Explain primary and secondary data
- Conduct simple surveys
- Design questionnaires
- Identify sources of statistical data

METHODS OF DATA COLLECTION

Data collection is the process of gathering information to be used for analysis. There are various methods to collect data depending on what information is needed and how it will be used.

TYPES OF DATA SOURCES

1. PRIMARY DATA

Primary data is information collected directly by the person who will use it. It is first-hand information collected for a specific purpose.

Characteristics:

- Original data
- Collected for a specific purpose
- More reliable and accurate
- Takes more time and effort to collect
- Can be expensive

2. SECONDARY DATA

Secondary data is information that has already been collected by someone else and is being reused.

Characteristics:

- Not original
- Collected for a different purpose
- Readily available
- Less expensive and faster to obtain
- May not fit your exact needs

COMPARISON: PRIMARY vs. SECONDARY DATA

Aspect	**Primary Data**	**Secondary Data**
Source	Original collection	Already exists
Cost	More expensive	Less expensive
Time	Takes longer	Faster to obtain
Accuracy	More reliable	May have errors
Suitability	Fits exact needs	May not fit perfectly
Control	Full control	No control
Examples	Own survey, experiment	Census data, books

EVALUATION

1. What is primary data? Give two examples
2. List three methods of collecting primary data
3. What is secondary data? Give two sources
4. State two advantages of using questionnaires

5. Why is sampling important in data collection?

ASSIGNMENT

1. Explain the difference between primary and secondary data
2. Describe how you would conduct an interview to collect data
3. Give three examples of secondary data sources in Nigeria
4. Design a simple 5-question questionnaire to find out students' favorite foods
5. What are the advantages and disadvantages of observation as a data collection method?

WEEK 10: EVERYDAY STATISTICS III

Learning Objectives

By the end of this lesson, students should be able to:

- Organize data in frequency tables
- Present data using bar charts
- Present data using pie charts
- Read and interpret charts
- Choose appropriate charts for different types of data

DATA PRESENTATION

After collecting data, it must be organized and presented in a way that is easy to understand. Raw data (unorganized data) is difficult to interpret, so we use tables and charts.

FREQUENCY TABLES

A frequency table is a way of organizing data to show how often each value occurs.

Key Terms:

FREQUENCY: The number of times a value or category appears in the data

TALLY: A way of counting using marks (|||| for 5)

Example 1: Creating a Frequency Table

Raw Data: Favorite fruits of 20 students Orange, Apple, Mango, Orange, Banana, Mango, Orange, Apple, Orange, Mango, Banana, Orange, Mango, Mango, Orange, Apple, Banana, Orange, Mango, Orange

Frequency Table:

Fruit	Tally	Frequency
Orange		
Mango		
Apple		
Banana		
Total		**20**

Example 2: Test Scores

Raw Data: Scores out of 10 7, 8, 6, 9, 7, 8, 10, 6, 7, 8, 9, 7, 6, 8, 7, 9, 8, 7, 10, 8

Frequency Table:

Score	Tally	Frequency
6		
7		
8		
9		
10		
Total		**20**

GROUPED FREQUENCY TABLES

When data has many different values, we group them into classes (intervals).

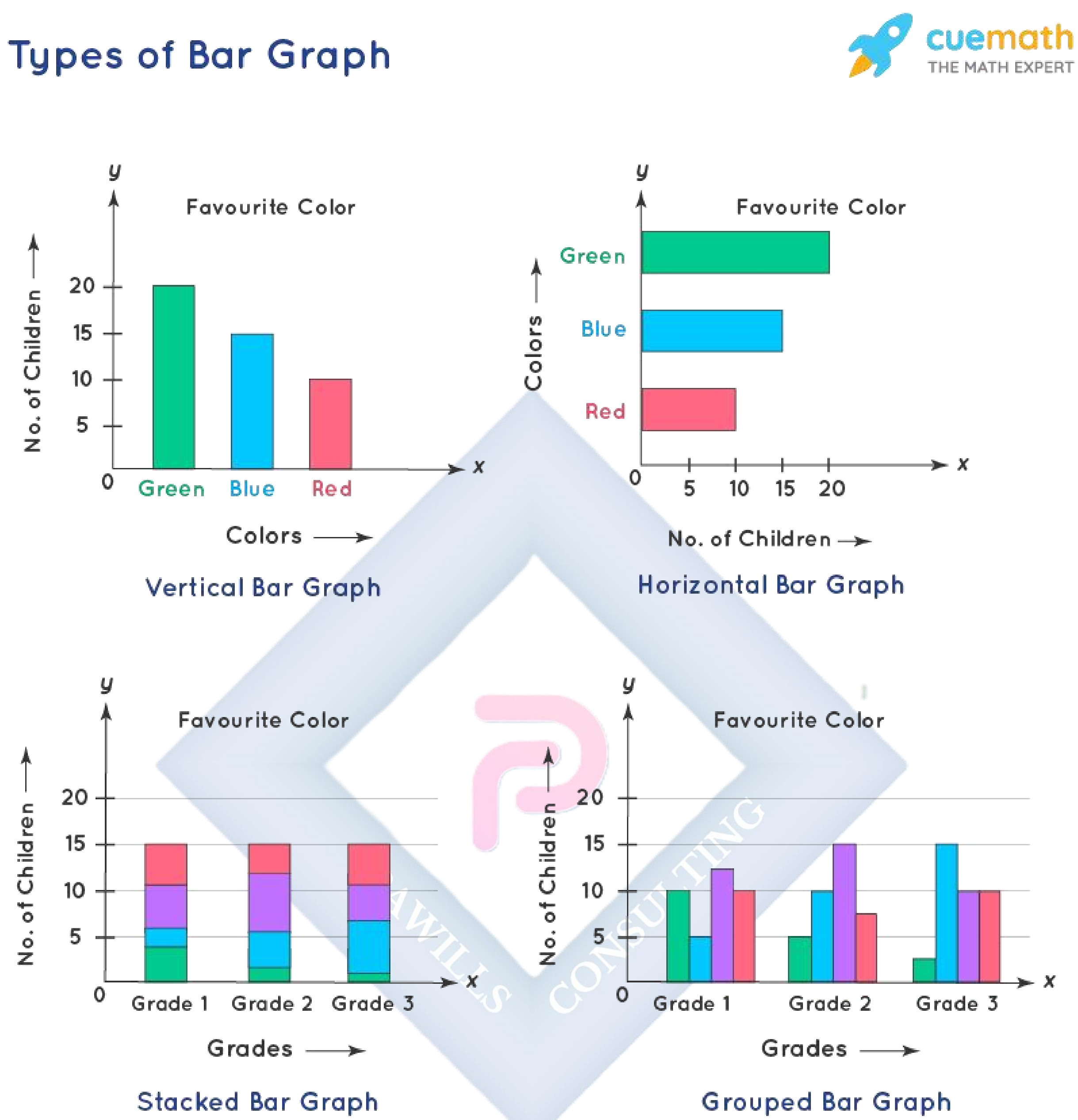
Example: Ages of 30 People

Raw Data: 12, 25, 34, 18, 45, 23, 29, 15, 38, 42, 27, 16, 31, 48, 22, 35, 19, 26, 33, 41, 17, 28, 36, 21, 30, 44, 20, 37, 24, 39

Grouped Frequency Table:

Age Group	**Tally**	**Frequency**
10-19		
20-29		
30-39		
40-49		
Total		**30**

BAR CHARTS



A bar chart uses rectangular bars to show data. The height of each bar represents the frequency.

Types of Bar Charts:

1. VERTICAL BAR CHART

- Bars go up (vertical)
- Categories on horizontal axis (x-axis)

- Frequency on vertical axis (y-axis)

2. HORIZONTAL BAR CHART

- Bars go across (horizontal)
- Categories on vertical axis
- Frequency on horizontal axis

Features of a Good Bar Chart:

1. **Title:** Clearly states what the chart shows
2. **Labels:** Both axes must be labeled
3. **Scale:** Uniform spacing on the frequency axis
4. **Bars:** Equal width, with gaps between them
5. **Colors:** Can use different colors for clarity (optional)

Example 1: Drawing a Vertical Bar Chart

Data: Favorite Sports of 50 Students

Sport	**Frequency**
Football	20
Basketball	12
Tennis	10
Athletics	8

Steps to Draw:

1. Draw two axes (horizontal and vertical)
2. Label horizontal axis "Sports"

3. Label vertical axis "Number of Students"
4. Write a scale on vertical axis (0, 5, 10, 15, 20...)
5. Draw bars for each sport with heights matching their frequency
6. Add title: "Favorite Sports of 50 Students"

Reading the Chart:

- Football is the most popular sport (highest bar)
- Athletics is the least popular (shortest bar)
- 20 students chose football
- Total students = 20 + 12 + 10 + 8 = 50

Example 2: Transportation to School

Method	Number of Students
Walk	15
Bus	25
Car	8
Bicycle	2

Total: 50 students

From this bar chart, we can see:

- Most students come by bus
- Very few students cycle to school
- 15 students walk to school

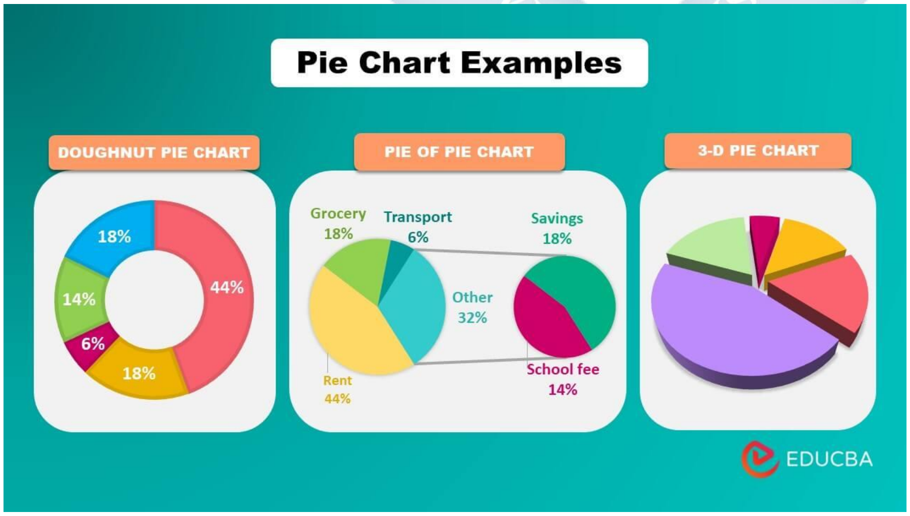
Example 3: Book Sales in a Week

Day	Books Sold
Monday	12
Tuesday	18
Wednesday	15
Thursday	22
Friday	30
Saturday	35

From this chart, we can observe:

- Saturday had the highest sales
- Monday had the lowest sales
- Sales generally increased towards the weekend
- Total books sold = 132

PIE CHARTS



A pie chart is a circular chart divided into sectors (slices). Each sector represents a category, and its size shows the proportion or percentage of that category.

Why called "Pie Chart"?

- It looks like a pie cut into slices
- Each slice represents part of the whole

When to Use Pie Charts:

- To show parts of a whole
- To show percentages
- When you want to compare proportions

Important Points:

- The whole circle = 360°
- Each sector angle = (Frequency/Total) × 360°
- All sectors together = 100%

Steps to Draw a Pie Chart:

1. Calculate total frequency
2. Find the angle for each sector: (Frequency ÷ Total) × 360°
3. Draw a circle
4. Mark the center
5. Draw sectors using a protractor
6. Label each sector
7. Add title

Example 1: Favorite Subjects

Data:

Subject	Number of Students
Mathematics	12

English	15
Science	9
Social Studies	6
Total	**42**

Step 1: Calculate angles

Mathematics: (12/42) × 360° = 102.86° ≈ 103° English: (15/42) × 360° = 128.57° ≈ 129° Science: (9/42) × 360° = 77.14° ≈ 77° Social Studies: (6/42) × 360° = 51.43° ≈ 51°

Total: 103° + 129° + 77° + 51° = 360° ✓

Step 2: Draw the circle and sectors

Using a compass, draw a circle. Using a protractor, draw each sector with its calculated angle.

Step 3: Label and add title

Label each sector with the subject name. Add title: "Favorite Subjects of 42 Students"

Example 2: Family Budget

Data:

Category	**Amount (₦)**
Food	40,000
Rent	30,000
Transport	15,000
Others	15,000
Total	**100,000**

Calculate angles:

Food: (40,000/100,000) × 360° = 144° Rent: (30,000/100,000) × 360° = 108° Transport: (15,000/100,000) × 360° = 54° Others: (15,000/100,000) × 360° = 54°

From this pie chart:

- Food takes the largest portion of the budget (40%)
- Rent is the second largest expense (30%)
- Transport and Others are equal (15% each)

Example 3: How Students Spend Free Time

Activity	Frequency	Angle
Playing	18	(18/60) × 360° = 108°
Reading	12	(12/60) × 360° = 72°
Watching TV	15	(15/60) × 360° = 90°
Sleeping	10	(10/60) × 360° = 60°
Others	5	(5/60) × 360° = 30°
Total	**60**	**360°**

INTERPRETING CHARTS

Reading Bar Charts:

Example: From a bar chart showing monthly rainfall:

- Which month had the highest rainfall? (Look for tallest bar)
- Which month had the lowest rainfall? (Look for shortest bar)
- What was the rainfall in June? (Read the height of June's bar)
- What is the total rainfall for the year? (Add all bars)

Reading Pie Charts:

Example: From a pie chart showing exam grades:

- Which grade was most common? (Look for largest sector)
- What percentage got grade A? (Read the label or calculate from angle)
- If there were 80 students total and grade B has angle 90°, how many got grade B?
  - 90°/360° = 1/4 = 25%
  - 25% of 80 = 20 students

PRACTICAL ACTIVITY

Activity: Survey and Create Charts

Task: Find out the favorite color of students in your class

Steps:

1. Ask each student their favorite color
2. Record using tally marks in a table
3. Count the frequencies
4. Calculate angles for pie chart
5. Draw both a bar chart and pie chart
6. Compare which chart shows the information better

Example Result:

Color	Frequency	Angle for Pie Chart
Blue	10	100°
Red	8	80°
Green	6	60°
Yellow	5	50°
Others	7	70°

Total	**36**	**360°**

EVALUATION

1. What is a frequency table?
2. Draw a bar chart for this data: Apples-15, Oranges-20, Bananas-12
3. Calculate the angle for a pie chart sector if 8 out of 32 students chose that option
4. When is it better to use a pie chart instead of a bar chart?
5. What must every bar chart have to be complete?

ASSIGNMENT

1. Create a frequency table for this data: Red, Blue, Red, Green, Blue, Red, Blue, Green, Red, Blue
2. If 12 out of 48 students chose Mathematics, what angle would it occupy in a pie chart?
3. Draw a simple bar chart showing: Dogs-6, Cats-4, Birds-8, Fish-2
4. List three features of a good bar chart
5. A pie chart sector has an angle of 90°. What fraction and percentage does it represent?

REVISION TIPS (WEEKS 11-13)

Key Topics to Review

1. **Simple Equations:** Solving one-variable linear equations
2. **Plane Shapes:** Properties of triangles, quadrilaterals, and circles
3. **3D Figures:** Cubes, cuboids, cylinders, spheres, cones - properties and formulas
4. **Constructions:** Using ruler and compass
5. **Angles:** Types, properties, angle sum in triangles and quadrilaterals
6. **Statistics:** Need for statistics, data collection methods, data presentation

Study Strategies

- **Practice solving equations daily** - at least 10 problems
- **Draw and label diagrams** for all plane and 3D shapes
- **Practice constructions** with actual ruler and compass
- **Memorize all formulas** for areas, volumes, perimeters
- **Create flashcards** for angle properties
- **Practice reading and drawing charts** - both bar and pie charts
- **Work in groups** to discuss angle problems
- **Time yourself** when solving problems
- **Check your answers** by substituting back
- **Make summary sheets** for each topic

IMPORTANT FORMULAS TO REMEMBER

Plane Shapes:

- Triangle area = ½ × base × height

- Rectangle area = length × width
- Square area = $side^2$
- Circle area = πr^2
- Circle circumference = $2\pi r$ or πd
- Perimeter = sum of all sides

3D Shapes:

- Cube volume = s^3
- Cuboid volume = $l \times w \times h$
- Cylinder volume = $\pi r^2 h$
- Sphere volume = $4/3\ \pi r^3$
- Cone volume = $1/3\ \pi r^2 h$
- Cube surface area = $6s^2$

Angles:

- Sum of angles in triangle = 180°
- Sum of angles in quadrilateral = 360°
- Angles on a straight line = 180°
- Angles at a point = 360°
- Complementary angles = 90°
- Supplementary angles = 180°

Statistics:

- Pie chart angle = (Frequency/Total) × 360°
- Percentage = (Part/Whole) × 100%

COMPREHENSIVE REVISION QUESTIONS

Simple Equations (5 questions)

1. Solve: $x + 15 = 42$
2. Find y if $3y = 51$
3. Solve: $2m - 7 = 19$
4. Calculate: $x/4 = 9$
5. Solve: $5n + 8 = 43$

Plane Shapes (5 questions)

6. Find the third angle of a triangle if two angles are 65° and 48°
7. Calculate the area of a rectangle 18 cm by 12 cm
8. A square has perimeter 48 cm. Find its area
9. Find the circumference of a circle with radius 14 cm ($\pi = 22/7$)
10. A trapezium has parallel sides 12 cm and 8 cm, and height 5 cm. Find its area

3D Shapes (5 questions)

11. Calculate the volume of a cube with side 6 cm
12. A cuboid is 10 cm long, 8 cm wide, and 5 cm high. Find its volume
13. How many faces does a cylinder have?
14. Find the surface area of a cube with edge 5 cm
15. A sphere has radius 21 cm. Find its surface area ($\pi = 22/7$)

Constructions (5 questions)

16. Name the instrument used to draw circles
17. Describe how to bisect a line segment
18. How do you construct a 60° angle using a compass?
19. What is the purpose of a protractor?
20. How can you construct a 45° angle?

Angles (10 questions)

21. Find the complement of 28°
22. Find the supplement of 115°
23. Two angles of a triangle are 72° and 58°. Find the third angle
24. In a quadrilateral, three angles are 85°, 90°, and 105°. Find the fourth angle
25. Classify an angle of 164°
26. If two complementary angles are in ratio 1:4, find the angles
27. In a parallelogram, one angle is 68°. Find the adjacent angle
28. Find the value of x if 3x + 2x + x = 180°
29. Each base angle of an isosceles triangle is 52°. Find the vertex angle
30. Two straight lines intersect. If one angle is 73°, find the other three angles

Statistics (10 questions)

31. What is the difference between primary and secondary data?
32. Give two methods of collecting primary data
33. A class has 40 students. In a survey about favorite subjects: Math-12, English-10, Science-14, Others-4. Create a frequency table
34. Calculate the angle for Mathematics in question 33 for a pie chart
35. Name three sources of secondary data
36. What is a questionnaire?
37. Why do we use sampling in data collection?
38. When is a bar chart more appropriate than a pie chart?
39. In a pie chart, if a sector has angle 72°, what percentage does it represent?
40. List three features of a good bar chart

Mixed Practice Problems

Problem 1: A rectangle has length 15 cm and width 9 cm. a) Find its perimeter b) Find its area c) If you increase each dimension by 3 cm, what is the new area?

Problem 2: Solve these equations: a) x - 18 = 25 b) 4y = 56 c) 3m + 7 = 28 d) n/6 = 7

Problem 3: A cylindrical water tank has radius 7 m and height 10 m. a) Calculate its volume (use π = 22/7) b) How many cubic meters of water can it hold? c) If water costs ₦50 per cubic meter, how much to fill the tank?

Problem 4: In a school survey of 60 students about transportation:

- 24 students walk
- 18 take the bus
- 12 come by car
- 6 use bicycles

a) Create a frequency table b) Calculate the angle for each category in a pie chart c) Draw a bar chart to represent this data d) Which is the most common method of transportation?

Problem 5: In triangle ABC:

- $\angle A = 3x$
- $\angle B = 2x$
- $\angle C = x$

Find: a) The value of x b) Each angle c) What type of triangle is this?

Problem 6: A cube has volume 512 cm^3. a) Find the length of its side b) Calculate its surface area c) How many edges does it have?

Problem 7: Construct: a) A line segment AB = 8 cm b) Bisect line AB c) Construct a perpendicular to AB at its midpoint d) Measure the perpendicular segments created

Problem 8: Four angles meet at a point. Three of them are 85°, 92°, and 108°. a) Find the fourth angle b) If two of these angles were equal, what would their sum be?

Problem 9: A school collected data on students' favorite fruits:

Fruit	**Number of Students**
Mango	45
Orange	30
Apple	15
Banana	30

a) How many students were surveyed? b) Calculate the angle for each fruit in a pie chart c) What percentage prefer mangoes? d) Draw a bar chart

Problem 10: A cone has:

- Radius = 6 cm
- Height = 8 cm

Find: a) The slant height (use Pythagoras: $l^2 = r^2 + h^2$) b) The volume (use $\pi = 3.14$)

QUICK REFERENCE GUIDE

Angle Facts:

- Acute: < 90°
- Right: = 90°
- Obtuse: 90° to 180°
- Straight: = 180°
- Reflex: 180° to 360°
- Complete: = 360°

Triangle Types:

- **By sides:** Equilateral (all equal), Isosceles (2 equal), Scalene (all different)
- **By angles:** Acute (all acute), Right (one 90°), Obtuse (one obtuse)

Quadrilateral Checklist:

- Square: 4 equal sides, 4 right angles
- Rectangle: Opposite sides equal, 4 right angles
- Parallelogram: Opposite sides equal and parallel

- Rhombus: 4 equal sides, opposite angles equal
- Trapezium: One pair of parallel sides

3D Shapes Summary:

- Cube: 6 faces, 12 edges, 8 vertices (all faces are squares)
- Cuboid: 6 faces, 12 edges, 8 vertices (all faces are rectangles)
- Cylinder: 2 circular faces + 1 curved surface
- Sphere: 1 curved surface
- Cone: 1 circular face + 1 curved surface + 1 vertex

ANSWERS TO SELECTED REVISION QUESTIONS

Simple Equations:

1. $x = 27$
2. $y = 17$
3. $m = 13$
4. $x = 36$
5. $n = 7$

Plane Shapes:

6. 67°

7. 216 cm²

8. 144 cm²

9. 88 cm

10. 50 cm²

3D Shapes:

11. 216 cm^3

12. 400 cm^3

13. 2 flat faces (both circular) + 1 curved surface = 3 surfaces total

14. 150 cm^2

15. 5,544 cm^2

Angles:

21. 62°

22. 65°

23. 50°

24. 80°

25. Obtuse angle

26. 18° and 72°

27. 112°

28. x = 30°

29. 76°

30. Vertically opposite = 73°; Adjacent angles = 107° (each)

Statistics:

31. Primary data is collected firsthand; secondary data already exists

32. Observation, questionnaire, interview, experiment, survey (any two)

33. Frequency table should show: Math-12, English-10, Science-14, Others-4, Total-40

34. (12/40) × 360° = 108°

35. Books, internet, government publications, newspapers, research papers (any three)

36. A set of written questions given to people to answer

37. When the population is too large to study completely

38. When comparing different categories or showing exact values

39. (72/360) × 100% = 20%

40. Title, labeled axes, uniform scale, equal width bars, gaps between bars (any three)